

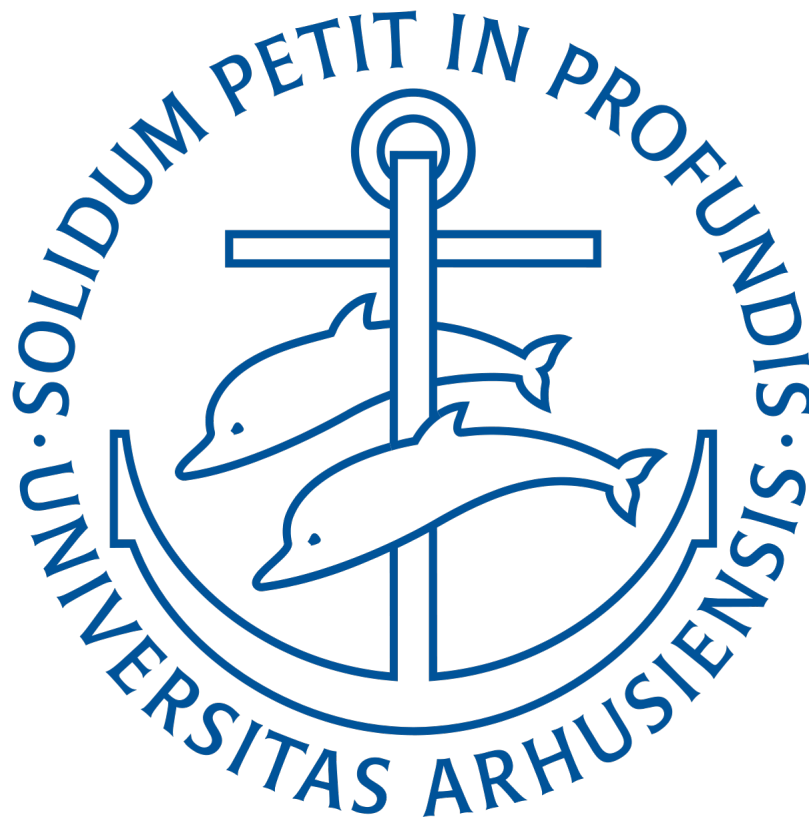
Deep learning model for fragment-level ctDNA detection

Deep learning model til detektering af ctDNA på fragment-niveau

Master thesis in data science

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Abstract

Early-stage cancer detection has been associated with substantially improved prognosis across multiple cancer types. However, effective screening strategies are not available for most cancers. Consequently, early-stage detection remains one of the most important challenges in cancer diagnostics. Circulating tumor DNA (ctDNA) has emerged as one of the most promising biomarkers in liquid biopsy for early-stage cancer detection. In particular, fragmentomic-based ctDNA detection methods have shown strong potential. Recent fragmentomic approaches have primarily relied on sample-level summary representations of cell-free DNA (cfDNA) fragmentation patterns, consequently leaving fragment-level ctDNA detection relatively unexplored. This thesis investigates whether deep learning models can identify ctDNA at the level of individual cfDNA fragments using a limited set of fragmentomic features.

Fragment length and end motifs were selected as the implemented cfDNA fragmentomic features. Multiple deep learning architectures, including a Feedforward Neural Network (FNN), Convolutional Neural Network (CNN), Bidirectional Long Short-Term Memory (BI-LSTM), and Bidirectional Gated Recurrent Unit (BI-GRU), were developed and evaluated. The models were optimized using Bayesian hyperparameter optimization and compared against a logistic regression baseline. To support a clinically relevant evaluation of the models, their ctDNA detection performance was evaluated at both fragment-level and sample-level. Finally, a SHapley Additive exPlanations (SHAP) analysis was used to investigate how the fragmentomic features influenced model predictions.

All deep learning architectures outperformed the logistic regression baseline in both the fragment-level and sample-level evaluation, indicating that nonlinear deep learning models can capture ctDNA predictive patterns beyond those available to conventional linear machine learning methods. However, only moderate fragment-level predictive performance was achieved across all architectures, which highlights the difficulty of reliable classification of individual fragments. The results further showed that decisions related to feature representation and integration had a greater influence on fragment-level predictive performance than the choice of sequence-modeling architecture. The performance differences observed between the architectures at fragment-level were relatively small. However, more pronounced differences emerged in the sample-level evaluation. In this setting, the CNN architecture achieved the strongest performance, particularly for samples containing low ctDNA fractions. The SHAP analysis further identified fragment length as the dominant predictive feature while demonstrating that end motif information provided complementary predictive value.

Overall, the findings suggest that fragment-level modeling represents a promising direction within fragmentomic-based ctDNA detection. While reliable classification of individual fragments remains challenging, the results indicate that aggregation of fragment-level predictions may facilitate viable sample-level ctDNA detection in early-stage cancer. Future work should focus on validation using independent clinical datasets.

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1 Introduction

Despite significant advances in the field of oncology, cancer still remains the second most common cause of premature deaths worldwide and is estimated to make number one by 2060 with an approximated 18.63 million deaths that year [1, 2]. Previous research has shown that the chances of surviving cancer are significantly improved if it is diagnosed and treated at an early clinical stage [3, 4]. However, approximately 50% of cancers are still only detected in advanced stages and most cancers do not have effective screening strategies [3, 5]. Traditionally, cancer detection has relied on a combination of tissue biopsy and a spectrum of imaging methodologies such as computerized tomography (CT) scans, magnetic resonance imaging (MRI), etc. Despite their widespread popularity, these techniques can only detect cancer when there is a noticeable anomaly in the tissue. When such anomalies become apparent, the cancer may already have evolved to an advanced stage or even metastasized [6, 7]. Furthermore, tissue biopsy is an invasive procedure associated with potential complications, while both tissue biopsy and imaging-based diagnostic techniques are costly and often limited in availability. Collectively, these limitations highlight the urgent need for the development of effective, scalable, and minimally invasive methods for early-stage cancer detection [8, 9, 7]. One such promising approach that may address these challenges is liquid biopsy. Liquid biopsy is the process of analyzing a sample of biological fluid, such as blood or urine, to detect and monitor various cancer related molecular biomarkers including but not limited to circulating tumor DNA (ctDNA) [7, 10]. A critical advantage of liquid biopsy is its potential to detect cancer at an early stage before it becomes symptomatic or visible on imaging tests. This potential arises from the fact that tumor cells release some biomarkers into the blood stream even when the tumor is small and localized [10]. Furthermore, liquid biopsy is a minimally invasive and cost-effective approach, making it a strong candidate for large-scale screening programs [10]. Consequently, recent years have seen extensive research into the diagnostic potential of various biomarkers in liquid biopsy. Despite several promising proof-of-concept studies, no method has yet achieved a detection level sufficient for routine implementation in screening programs, highlighting the need for further research in this field [10, 11].

Among the various biomarkers investigated in liquid biopsy, ctDNA is considered one of the most promising and is therefore selected as the primary focus of this thesis [11, 12, 13]. To understand the role of ctDNA as a biomarker, it is first necessary to introduce the concept of cell-free DNA (cfDNA). In eukaryotic cells, DNA is wrapped around histone protein complexes, known as nucleosomes. This gives the DNA a characteristic “beads-on-a-string” appearance, as shown in Figure 1b. When cells undergo apoptosis, necrosis, or active secretion, their DNA is fragmented and released into the bloodstream. These circulating DNA fragments are referred to as cfDNA. Fragmentation typically occurs in the linker DNA between nucleosomes, resulting in fragments that predominantly correspond to mononucleosomes (i.e. contains one nucleosome), with a characteristic length of approximately 167 base pairs (bp), and a small fraction corresponding to dinucleosomes (i.e. contains two nucleosomes), with lengths of approximately 330 bp [5, 14]. This fragmentation process is shown in Figure 1c. When tumor cells undergo apoptosis, necrosis, or active secretion, their DNA is similarly fragmented and released into the bloodstream. These circulating tumor DNA fragments are referred to as ctDNA. Consequently, in individuals with cancer, ctDNA constitutes a fraction of the total cfDNA in circulation [15]. Monitoring ctDNA levels in an individual can therefore serve as a biomarker for cancer, as ctDNA originates from tumor cells. However, a central

challenge lies in distinguishing ctDNA fragments from the background of cfDNA released by normal cells. This challenge is compounded by the low abundance of ctDNA in early-stage cancer, where it may constitute as little as $< 0.1\%$ of total cfDNA [16]. Nevertheless, numerous studies have focused on developing methods to identify ctDNA based on tumor-specific molecular features, including somatic mutations, epigenetic alterations, and characteristic fragmentation patterns. Among these, approaches based on somatic mutations have historically received the most attention. These approaches initially gained traction in the context of patient monitoring during therapy, where the patient’s tumor-specific mutations are known and can be used to reliably monitor ctDNA levels [5, 12]. However, in early-stage cancer detection, the mutational profile of a patient’s tumor is not known a priori and therefore cannot be used for targeted monitoring of ctDNA. Instead, panels of genes frequently mutated across cancers are interrogated for the presence of mutations in cfDNA fragments to detect evidence of ctDNA. However, non-tumor-derived sources of somatic variation, such as mutations in white blood cells, may also be detectable in cfDNA and can confound this analysis, thereby reducing the specificity of mutation-based approaches [5]. Furthermore, the large size of the human genome presents an additional challenge for mutation-based ctDNA detection. Although ctDNA originates from tumor cells, only a small fraction of genomic positions in a tumor cell typically harbor somatic mutations. As a result, the majority of ctDNA fragments in circulation do not contain detectable mutations. Consequently, approaches relying solely on mutation detection are limited to a small subset of ctDNA fragments, further reducing sensitivity. This is particularly critical in early-stage cancer detection where ctDNA levels are already low [14]. As a result, recent research has increasingly focused on identifying novel tumor-specific molecular features to improve the sensitivity of ctDNA detection in circulation. In this context, approaches based on characteristic fragmentation patterns (i.e. fragmentomics) have emerged as a promising strategy.

Several studies have demonstrated that ctDNA fragments tend to be shorter and exhibit greater diversity in the terminal nucleotides (i.e. end motifs) than non-tumor-derived cfDNA fragments [5, 14, 17, 18, 19, 20]. In addition, certain genomic positions have been shown to be more frequently associated with fragment ends than others, commonly referred to as preferred ends, and these preferred end patterns tend to differ between ctDNA and non-tumor-derived cfDNA fragments. Furthermore, differences in nucleosome occupancy have also been reported between tumor and non-tumor cells. As nucleosomes protect DNA from fragmentation, genomic regions occupied by nucleosomes are more likely to be covered by cfDNA fragments rather than contain fragment ends, resulting in characteristic coverage patterns across the genome. Since nucleosome occupancy differs between tumor and non-tumor cells, these coverage patterns are also expected to differ between ctDNA and non-tumor-derived cfDNA [21, 22]. These characteristic differences between fragmentomic features from ctDNA and cfDNA fragments have led to increasing interest in fragmentomics as a potential approach for ctDNA detection. Figure 1a shows the underlying processes of extracting these fragmentomic features from cfDNA fragments.

Recent studies have primarily relied on sample-level summary representations of fragmentomic features to infer the presence of ctDNA at sample-level. Such representations may take the form of fragment length distributions or end motif frequency profiles derived from all cfDNA fragments in the sample. Because fragmentomic features differ between ctDNA and non-tumor-derived cfDNA fragments, the resulting summary representations are expected to differ between samples containing ctDNA and samples composed exclusively of non-tumor-derived cfDNA. Consequently, the summary

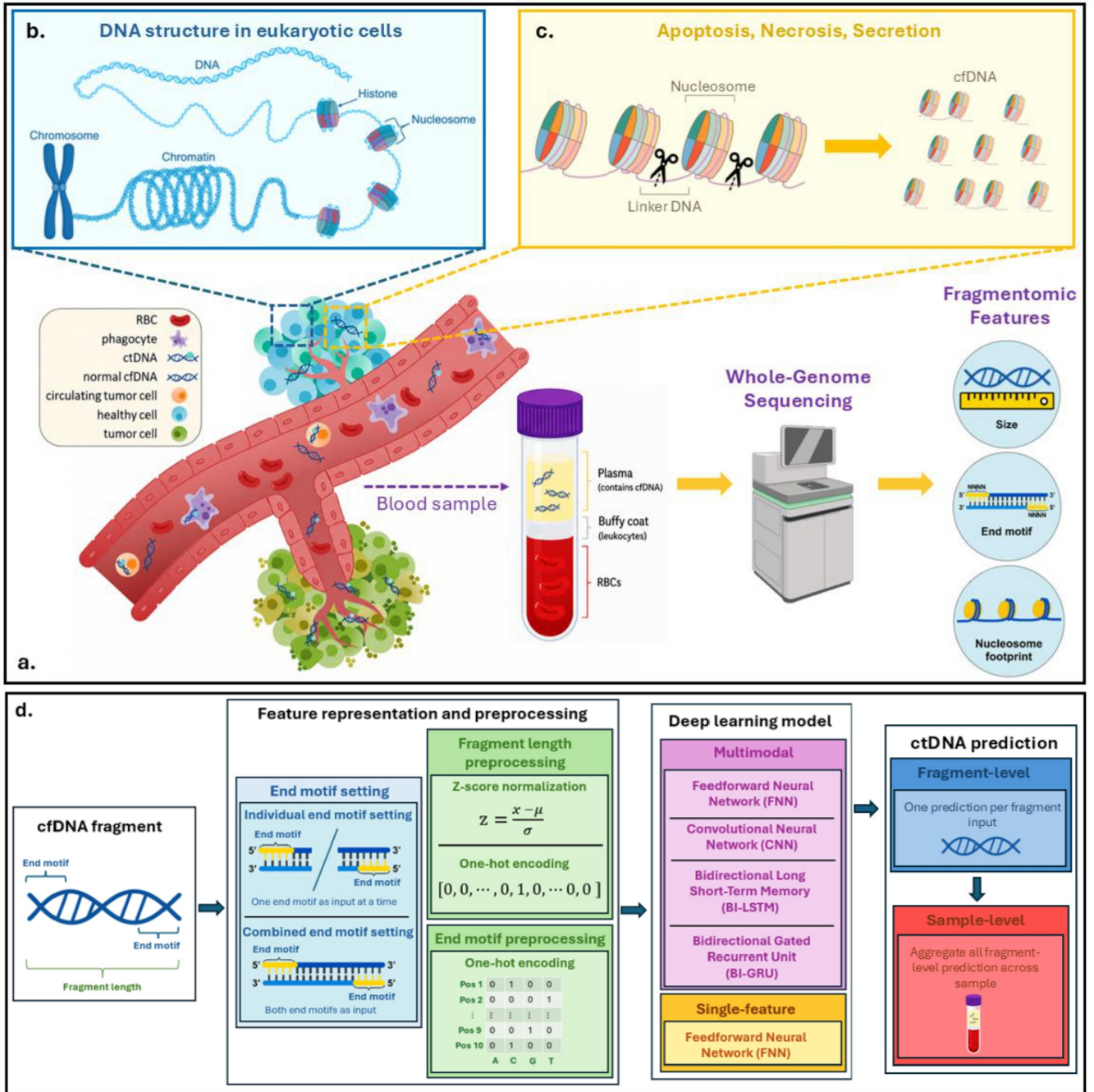


Figure 1: Schematic overview of the biological processes underlying cfDNA fragmentomics, the extraction of fragmentomic features, and the predictive framework introduced in this thesis. **a.** Illustration of the biological processes leading to the release of cfDNA into the bloodstream and the extraction of fragmentomic features from a blood sample. **b.** DNA organization in eukaryotic cells illustrating the “beads-on-a-string” structure formed by DNA wrapped around nucleosomes. **c.** DNA fragmentation processes, including apoptosis, necrosis, and active secretion, leading to the release of cfDNA fragments. **d.** Overview of the proposed fragment-level ctDNA prediction framework based on selected fragmentomic features (end motifs and fragment length). The framework includes multiple options for feature representation, feature preprocessing, and deep learning architectures. The figure is adapted from illustrations by Ding and Lo [23] and Wong [24].

representations can be used to distinguish between samples with and without ctDNA. However, when considering only the four terminal nucleotides of each fragment end, this results in 256 possible end motifs; a number that increases exponentially as additional nucleotides are included. Furthermore, current sequencing technologies enable the analysis of fragment lengths exceeding 500 base pairs. Consequently, representing fragmentomic features through summaries at this level of detail gives rise to high-dimensional data. Furthermore, although cfDNA is known to originate from processes such as apoptosis, necrosis, and active secretion, the precise biological mechanisms by which these processes influence cfDNA concentration and fragmentomics remain poorly understood. This limited understanding makes it challenging to accurately model and predict the biological processes governing ctDNA generation and behavior [14]. The combination of the high-dimensional nature of fragmentomic features and the limited understanding of the biological processes governing cfDNA generation strongly motivates the use of deep learning (DL) and machine learning (ML) approaches for fragmentomic-based ctDNA detection. Both DL and ML methods are particularly well suited for analyzing high-dimensional data and modeling complex and poorly understood patterns, as they can automatically learn intricate relationships directly from data [22]. An illustrative example of such an approach is the DNA Evaluation of Fragments for Early Interception (DELFI) algorithm. In this algorithm, Cristiano et al. [25] represented each sample using a genome-wide fragmentation profile generated by mapping cfDNA fragments to their genomic locations and assessing fragment lengths across 504 non-overlapping windows of 5 Mb, covering approximately 2.6 Gb of the genome. For each window, they calculated the ratio of short cfDNA fragments (100–150 bp) to longer fragments (151–220 bp), as well as overall coverage. These fragmentation profiles were then used to train a gradient boosting model for the detection of multiple cancer types. The method achieved sensitivities ranging from 57% to over 99% across seven cancer types, with an overall area under the curve (AUC) of 0.94, demonstrating a strong proof-of-concept performance [25]. Another example of such an approach is provided by Liu et al. [26], who developed an instruction-tuned large language model termed iLLMAC. In this model, the cfDNA end motif feature was encoded using multiple representations, including end motif “sentences” (i.e. sequences formed by concatenating end motifs ordered by frequency), fragmentation profiles derived from non-negative matrix factorization of end motif frequency histograms, and motif diversity scores capturing the distribution of end motif frequencies. The model achieved an AUC of 0.912 on an external validation cohort comprising multiple cancer types and non-cancer controls [26]. In both examples above, the methods rely on a single fragmentomic feature, i.e. end motifs or fragment length. However, approaches that integrate multiple fragmentomic features have recently gained increasing attention. One such example is the multimodal assay SPOT-MAS, introduced by Nguyen et al. [27]. SPOT-MAS is a machine learning framework combining methylomic and fragmentomic features (including fragment length and end motifs) for ctDNA detection. The method achieved sensitivities of 73.9% and 62.3% for stage I and II cancers, respectively, increasing to 88.3% for non-metastatic stage IIIA [27]. Despite these promising proofs of concept, no fragmentomic-based ctDNA detection method has yet been approved for clinical implementation. This indicates that further research is needed in order to fully unlock the potential of fragmentomic-based cancer detection [22].

All of the studies discussed above have relied on sample-level summary representations of the fragmentomic features. This reflects a broader trend within fragmentomic-based ctDNA detection and has consequently left fragment-level ctDNA detection relatively unexplored [5]. In a fragment-level setting, each individual cfDNA fragment is classified as either ctDNA or non-tumor-derived

cfDNA. In principle, a sufficiently reliable fragment-level classifier could enable ctDNA detection even when only a very limited number of ctDNA fragments are present within a sample. However, ctDNA detection in early-stage cancer has been proposed to be inherently challenging due to a low signal-to-noise ratio [28]. Consequently, it might be difficult for a fragment-level approach to achieve the level of reliability required for confident classification of individual fragments. Nevertheless, aggregating the predictions from a moderately accurate fragment-level classifier may still result in reliable sample-level classification. Fragment-level modeling therefore represents a promising and largely unexplored direction within fragmentomic-based liquid biopsy research. Furthermore, as exemplified by the SPOT-MAS study, an increasing number of novel fragmentomics-based methods are adopting multimodal approaches that integrate multiple fragmentomic and molecular features [22]. However, given the low signal-to-noise setting of ctDNA detection in early-stage cancer, simpler model architectures may be more effective than highly complex models, as they are less prone to overfitting noise. This may partly explain why, despite the theoretical advantages of multimodal approaches (i.e. enabling more comprehensive characterization of ctDNA signals), they often provide only incremental improvements in early-stage cancer detection compared to simpler models [22]. Consequently, adding more fragmentomic features does not necessarily improve performance but may instead introduce unnecessary model complexity.

These observations lead to the main goal of this thesis, which is *to develop and evaluate a series of models for fragment-level ctDNA detection using a limited set of fragmentomic features*. The thesis focuses particularly on whether relatively simple architectures are capable of effectively identifying ctDNA at the level of individual fragments while limiting unnecessary model complexity. Although the main goal of the thesis is to develop models for fragment-level ctDNA detection, a clinically relevant implementation of the models would ultimately require ctDNA detection at sample-level. Consequently, an evaluation framework would have to be developed to investigate whether the fragment-level predictions generated by the models can support accurate sample-level ctDNA classification.

Focusing on individual fragments rather than sample-level summary representations reduces the dimensionality of the fragmentomic data. However, depending on the selected preprocessing strategies, the problem remains relatively high-dimensional. Furthermore, fragment-level ctDNA detection is still governed by the same complex and poorly understood biological mechanisms as summary-level approaches. Consequently, the application of deep learning methods remains strongly motivated in the fragment-level setting. To investigate the objectives of this thesis, multiple deep learning architectures and data modeling strategies are therefore explored and compared.

The thesis is structured as follows. Section 2 describes the selected fragmentomic features, the datasets used for model training and evaluation, the architectures and their methodological considerations, and the procedures used for training and evaluating the models. Section 3 presents the results obtained from the training and evaluation of the models. Section 4 discusses these findings in relation to previous research and considers their implications for future studies in fragmentomic-based ctDNA detection. Finally, Section 5 concludes the project.

2 Materials and methods

In this section, the fragmentomic features selected for implementation in this thesis are first described. Subsequently, the datasets used for training and evaluation of the introduced models are presented, followed by a description of the preprocessing strategies applied to the selected fragmentomic features. Then the architectures of the implemented models are introduced together with the methodological considerations underlying their design. Finally, the procedures used for selecting, training and evaluating the models are described. An overview of the prediction framework presented throughout this section is shown in Figure 1d.

2.1 Data

2.1.1 Fragmentomic feature selection

As mentioned above, the primary objective of this thesis is to develop and evaluate models for fragment-level ctDNA detection using a limited set of fragmentomic features. To enable the development of such models, it is first necessary to determine which fragmentomic features should be included. Since only a limited set of fragmentomic features are to be included, the selection should focus on features with a well-established and reproducible signal. Recent research in fragmentomic-based ctDNA detection has primarily focused on four major fragmentomic features: fragment length, end motifs, nucleosome occupancy, and preferred ends, all of which were described in the introduction. Among these, fragment length and end motifs are the most extensively studied and have demonstrated strong reproducibility across multiple cancer types[21, 22]. Furthermore, fragment length and end motifs are also particularly well suited for fragment-level modeling, as both features are inherently defined at the level of individual fragments. In contrast, nucleosome occupancy and preferred ends are typically inferred from aggregated fragmentation patterns across multiple fragments rather than from individual fragments directly [21, 22]. Based on these considerations, fragment length and end motifs were selected as the fragmentomic features implemented in this thesis. Consequently, a dataset containing both fragment length and end motif information at fragment-level is required. In the following section, the dataset used in this thesis is described.

2.1.2 Dataset

The data described in the following section has previously been published by Frydendahl et al. [30]. It contains cfDNA fragments from 90 patients diagnosed with UICC stage III colorectal cancer, recruited from seven Danish hospitals between November 2002 and February 2019. All patients received treatment (i.e. surgery with or without adjuvant chemotherapy) and recurrence surveillance. Blood samples were collected before surgery, after surgery, and during follow-up visits every three months for up to three years. If a patient experienced recurrence, sampling continued until recurrence treatment began. The blood samples were centrifuged to separate the red blood cells (RBCs), the buffy coat (containing white blood cells) and the plasma (containing cfDNA). Furthermore, tumor tissue and paired buffy coat samples were collected from each patient. All tumor, buffy coat, and cfDNA samples underwent whole-genome sequencing (WGS). Comparison between the sequenced tumor tissue and paired buffy coat (i.e. normal DNA) enabled identification of somatic single nucleotide variants (SNVs) in the tumor. A further comparison between the buffy

coat and the cfDNA fragments enabled identification of somatic SNVs in the individual fragments. An intersection of the tumor-informed somatic SNVs and the cfDNA somatic SNVs was then used to identify cfDNA fragments spanning tumor-informed SNVs. Any cfDNA fragment spanning a tumor-informed SNV was classified as a ctDNA fragment. Consequently, if any blood sample contained a ctDNA classified fragment, the sample was in turn classified as ctDNA-positive. A total of 712 blood samples were collected with 508 classified as ctDNA-negative and 204 as ctDNA-positive. This sampling procedure is illustrated in Figure 2. The ctDNA classified fragments were combined with non-tumor derived cfDNA fragments to generate a dataset enabling fragment-level ctDNA detection. To ensure that the non-tumor derived cfDNA actually were non-tumor, they were only taken from follow-up samples from patients who had no ctDNA-positive samples throughout the entire follow-up period. The non-tumor derived cfDNA fragments were further chosen such that they had to span the location of a tumor-informed cfDNA somatic SNV. This resulted in a dataset containing 1.556.452 fragments with a ctDNA rate of approximately 58%. To enable model development and evaluation, the resulting dataset was divided into training, validation, and test subsets. The data was partitioned at patient level to ensure that no patient overlap occurred between the subsets. The dataset splits were determined based on a compromise between dataset size and ctDNA fraction. Approximately 20% of the total cfDNA fragments were assigned to the

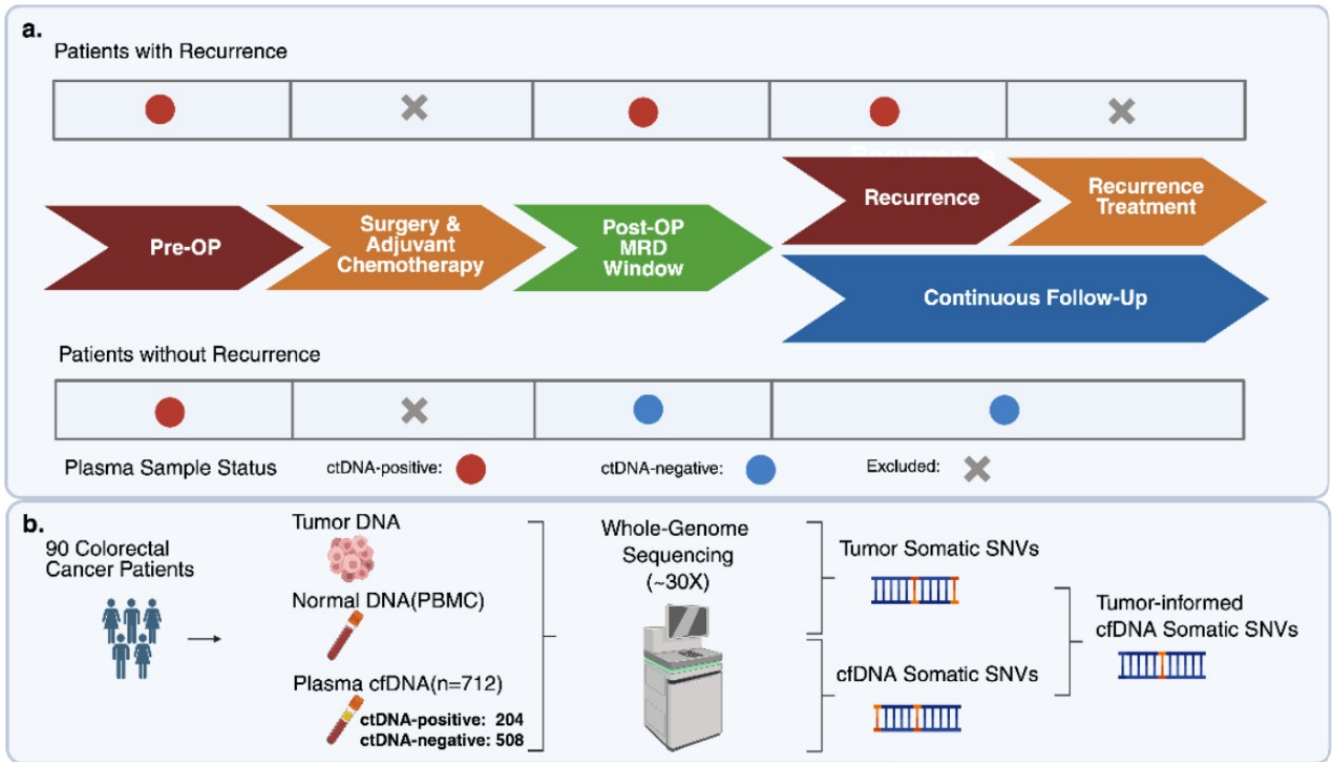


Figure 2: Systematic overview of the sampling procedure. **a.** Time points of cfDNA sample collection relative to clinical interventions, including pre-operative, surgery and adjuvant chemotherapy (if administered), and post-operative Minimal Residual Disease (MRD) window. **b.** Available patient data and sequencing workflow. Whole-genome sequencing of tumor tissue, paired buffy coat, and cfDNA samples, followed by somatic variant calling, resulted in the identification of tumor-informed cfDNA somatic SNVs. The figure is an adaptation of the figure made by Lin et al. [29].

test dataset, while approximately 20% of the remaining fragments were allocated to the validation dataset. Furthermore, all three subsets were constructed to maintain ctDNA fractions comparable to that of the original dataset.

For each fragment, the dataset contains the two selected fragmentomic features, namely fragment length and end motifs. Fragment length is defined as the number of nucleotide base pairs within the fragment and is therefore represented in the dataset as an integer. The fragment lengths range from 10 to 500 base pairs. Each end motif consists of the 10 terminal nucleotides at the corresponding fragment end and is represented as a string composed of the nucleobases adenine (A), thymine (T), cytosine (C), and guanine (G). The dataset further contains a sample ID, a patient ID and a binary ctDNA label per fragment.

2.1.3 Preprocessing

The next step is to determine how the selected fragmentomic features should be preprocessed before they are used as input to the models. The preprocessing strategies introduced in the following section are illustrated in Figure 1d.

As mentioned, in their raw form the fragment lengths are represented as integers, whereas the end motifs are represented as strings. Implementation of the fragment length feature is therefore relatively straightforward, as it is inherently numerical. However, the ctDNA signal associated with fragment length arises from relative deviations in the length distribution of ctDNA fragments compared to non-tumor-derived cfDNA fragments, rather than from the absolute fragment lengths themselves [17, 19]. Consequently, z-score normalization of the fragment lengths may be beneficial, as it allows the model to focus on relative differences in fragment size. Using z-score normalization would result in the fragment lengths being represented as a real number instead of an integer. An alternative preprocessing strategy is to represent fragment lengths using one-hot encoding, thereby treating fragment length as a categorical rather than continuous feature. This approach may be advantageous, as it reflects the discrete nature of fragment lengths, where the associated signal may vary substantially between neighboring lengths rather than exhibiting a smooth continuous transition [31]. Furthermore, one-hot encoding enables the model to learn complex interactions between specific fragment lengths and end motifs when both features are implemented. As mentioned above, the fragment length range from 10 to 500 base pairs resulting in 491 possible fragment lengths. Consequently, using one-hot encoding would result in the fragment lengths being represented through vectors of length 491.

Unlike fragment length, the end motif representations are not inherently numerical and therefore require preprocessing before they can be used as model inputs. Since the ctDNA signal associated with end motifs is encoded in both the specific nucleotide composition and the positional arrangement of the nucleotides, an appropriate preprocessing strategy should preserve both the categorical nature of the nucleotides and their positional information [20]. A strategy that achieves this is one-hot encoding. By one-hot encoding each nucleotide position independently, the positional information of the nucleotides is preserved while maintaining the categorical structure of the end motifs. As each nucleotide position can take four possible values, one-hot encoding them would result in a vector of length 4. Consequently, as each end motif contains 10 nucleotides the one-hot strategy results in a 10×4 matrix representation of the end motifs. Furthermore, as each fragment

has two corresponding fragments ends the end motifs for a single fragment are represented through two 10×4 matrices.

The effects of the aforementioned preprocessing techniques will be examined across all the different model architectures introduced in this thesis.

2.1.4 End motif representation

As both end motifs of a single fragment originate from the same biological process, they would be expected to express similar biological signals. Consequently, the prediction of whether a cfDNA fragment is ctDNA may not be strongly affected by whether one or both end motifs are included in the prediction. However, implementing only a single end motif in a fragment-level prediction model results in a simpler model architecture compared to implementing both end motifs. To investigate the predictive capability of this simpler architecture, a second version of the dataset is created in which each fragment is represented by two observations, one for each end motif, where only the end motif differs. This results in a dataset containing twice as many observations as the original dataset. Throughout the remainder of this thesis, these two representations are referred to as the *individual end motif setting* and the *combined end motif setting*, corresponding to whether an observation contains a single or both end motifs for the corresponding fragment, respectively. These two possible end motif settings are illustrated in Figure 1d. The effect of implementing either one or both end motifs per prediction is evaluated across all model architectures introduced in this thesis.

2.2 Model architecture

This section describes the different deep learning architectures introduced in this thesis and the methodological considerations underlying their design. An overview of these architectures is given in Figure 1d.

2.2.1 Feedforward Neural Network

As discussed in the introduction, the task of fragmentomic-based fragment-level ctDNA detection strongly motivates the application of deep learning architectures. Furthermore, since part of the objective of this thesis is to develop a fragment-level ctDNA detection model with minimal architectural complexity, the first architecture introduced is the simplest form of deep learning architecture, namely a Feedforward Neural Network (FNN). An FNN consists of a Neural Network with fully connected layers in which information propagates exclusively in the forward direction from the input layer toward the output layer. A further goal of the thesis is to limit the number of fragmentomic features implemented. Therefore, the FNN architecture is introduced in both a single-feature and multimodal version, based on the two selected fragmentomic features, enabling evaluation of a single-feature approach versus a multimodal approach. Below follows a description of the FNN architecture for the single-feature and multimodal approach.

Single-feature

To maintain a simple model structure, the FNN implemented for the single-feature approach contains only a single hidden layer with a Rectified Linear Unit (ReLU) activation function. Furthermore, as fragment-level ctDNA detection constitutes a binary classification task, the output layer consists of a single neuron. A sigmoid activation function is applied to the output of this neuron, thereby enabling the output value to be interpreted as the probability of a fragment being ctDNA. This general FNN architecture for the single-feature setting is illustrated in Figure 3. The architecture does however vary slightly between the individual and combined end motif setting when implemented for the end motif feature. In the individual end motif setting the single-feature FNN architecture directly follows that illustrated in Figure 3. However, the combined end motif setting requires the model to jointly process both end motifs associated with each fragment. This is done by passing each of the two end motifs through the first fully connected layer independently, generating the hidden layer with ReLU activation for each end motif. These hidden layers are then concatenated and passed through the last fully connected layer with a single output neuron, following the output structure as illustrated in Figure 3. The reason why the two end motif are passed through the same initial fully connected layer is because any important biological signal is expected to be expressed similarly in both end motifs.

The dimensionality of the input layer depends on the selected fragmentomic feature and preprocessing strategy. As mentioned above, two preprocessing strategies are considered for the fragment length, i.e. normalization and one-hot encoding. Here, fragment length is represented as a real number or a one-hot vector of length 491 when using normalization or one-hot encoding, respectively. Consequently, when implementing the FNN architecture exclusively for fragment length the input layer consists of either 1 or 491 neurons depending on the applied preprocessing. In contrast to fragment length, only a single preprocessing strategy is considered for the end motifs, namely one-hot encoding. Here, each end motif is represented as a 10×4 matrix. To enable its use as input to the FNN, this representation is flattened into a vector of length 40. In both the individual and combined end motif setting, the FNN architecture takes one end motif as input at a time. Consequently, when implementing the FNN architecture exclusively for end motifs, the input

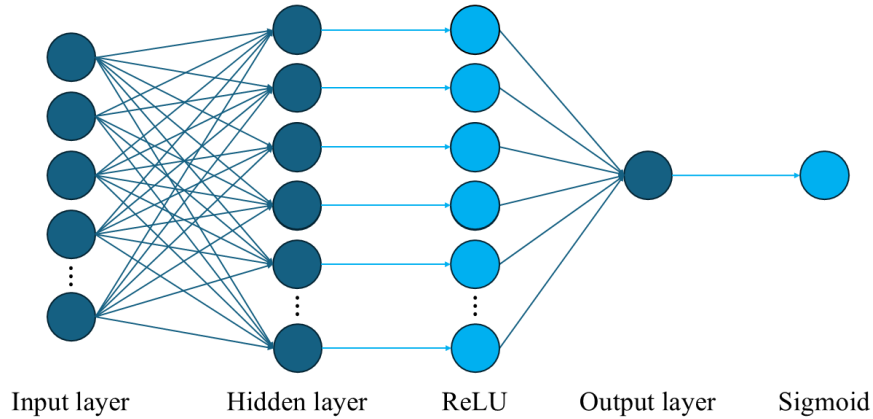


Figure 3: The single-feature FNN architecture.

layer consists of 40 neurons. The number of neurons in the hidden layer is determined through hyperparameter optimization.

Multimodal

In the multimodal approach, the FNN architecture is separated into three branches; a fragment length branch, an end motif branch, and a feature fusion branch. The fragment length branch models the preprocessed fragment length through a fully connected layer with ReLU activation. The end motif branch similarly models the flattened preprocessed end motif through a fully connected layer with ReLU activation. Finally, the feature fusion branch concatenates the features learned from the fragment length branch and end motif branch and passes them through a FNN with the same model architecture as the single-feature FNN. The general architecture for the multimodal FNN setting is illustrated in Figure 4. However, the architecture again varies slightly between the individual and combined end motif setting. In the individual end motif setting, the multimodal FNN architecture directly follows that illustrated in Figure 4. In contrast, the combined end motif setting again requires the model to jointly process both end motifs associated with each fragment. This is achieved by processing the two end motif independently through the end motif branch and subsequently concatenating the two feature representations from the branch. The concatenated representation is then passed to the feature fusion branch.

The dimensionality of the input layer of the fragment length and end motif branch follows that of the single-feature FNN implemented for the corresponding feature. The number of neurons in the hidden layers are determined through hyperparameter tuning.

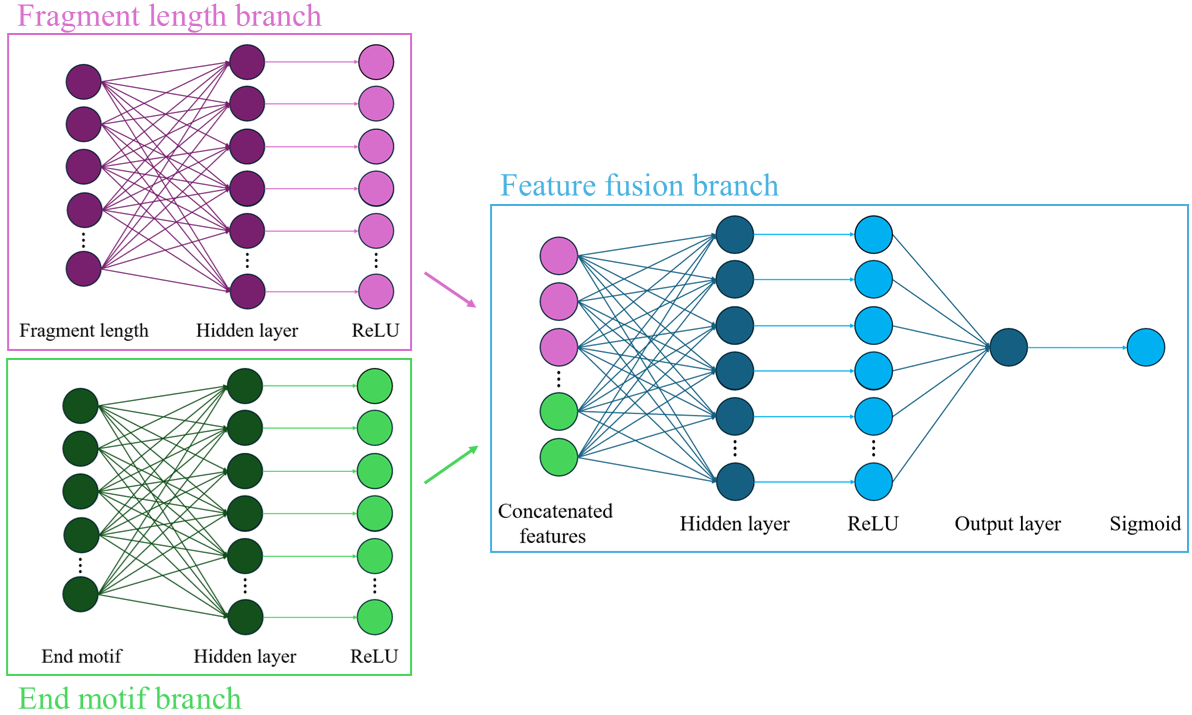


Figure 4: The multimodal FNN architecture.

2.2.2 Convolutional Neural Network

The primary objective of the FNN architecture described in the previous section was to maintain minimal model complexity. However, previous research in related sequence-based prediction tasks suggests that a moderate increase in architectural complexity may be beneficial for modeling the end motif sequences. In recent years, Recurrent Neural Networks (RNNs) and Transformer-based architectures have achieved considerable success across a wide range of genomic sequence-based prediction tasks, including in the field of ctDNA detection, as exemplified by models such as iLLMAC [26, 32]. This seems like a natural development when considering the strong performance of these architectures in natural language processing and other sequential modeling problems [32]. Nevertheless, the primary advantage of RNN and Transformer architectures lies in their ability to capture long-range sequence dependencies [32, 33]. While such dependencies may be relevant in summary-level ctDNA detection approaches, as in the case of iLLMAC, the fragment-level setting considered in this thesis involves end motifs of only 10 base pairs. Furthermore, Transformers constitute highly complex deep learning architectures [34, 35, 36]. Consequently, Transformer-based models may introduce unnecessary architectural complexity without providing substantial improvements in predictive performance when applied to prediction tasks involving relatively short DNA sequences [33]. In contrast, convolutional neural networks (CNNs) have demonstrated strong performance across multiple sequence-based prediction tasks involving short DNA sequences [32, 33]. Traditionally, CNNs were developed for computer vision applications and remain among the most successful deep learning architectures for image analysis due to their powerful feature extraction capabilities [32, 37]. More recently, however, CNNs have been successfully adapted to genomic applications by applying convolutional operations to numerically encoded DNA sequences. In this context, the feature extraction capabilities of CNNs have proven highly effective for identifying meaningful recurring sequence patterns, such as local genomic motifs, even under low sequencing coverage conditions [32, 33, 38]. This has led to the development of several successful CNN-based genomic models such as DeepBind, a single-layer CNN designed for protein-binding prediction, and DeepSEA, a three-layer CNN developed to predict the effects of noncoding variants [39, 40]. Notably, both DeepBind and DeepSEA employ relatively shallow CNN architectures. This observation is consistent with findings reported by Zeng et al. [41], who conducted a comprehensive analysis of the impact of CNN architecture on protein-binding prediction performance. Their study demonstrated that increasing network depth reduced motif discovery performance compared to simpler single-layer architectures. In contrast, increasing the number of convolutional kernels improved predictive performance, consistent with the common interpretation of convolutional kernels as motif detectors. The study ends up concluding that the performance of a CNN does not increase monotonically with their complexity and the design needs to match the learning task at hand [41]. These findings suggest that while the introduction of CNN-based feature extraction may be beneficial for genomic sequence-based prediction tasks, the overall architectural complexity should remain limited and be adjusted to fit the specific task.

Combined, these observations indicate that fragment end motifs may be more effectively modeled using a shallow CNN architecture rather than a FNN architecture. This forms the basis of the next model architecture, which builds on the multimodal FNN architecture from the previous section by incorporating a convolutional layer in the end motif branch. In this multimodal CNN architecture, the fragment length feature is still processed through a fully connected linear layer followed by a ReLU activation function. However, the end motifs are processed using a single one-dimensional

convolutional layer also with a ReLU activation. Based on the findings reported by Zeng et al. [41], the convolutional architecture is kept shallow and adapted specifically for the ctDNA detection task. The adaptations are made to preserve the positional information of the end motifs throughout the architecture, as this is expected to contain biologically relevant signals. To achieve this, padding is utilized such that the length of the convolutional feature maps matches the length of the input end motif. Furthermore, no pooling operations are applied, thereby preventing a reduction of the spatial resolution of the convolutional feature maps. The resulting feature maps are subsequently flattened into a vector before being passed to the feature fusion branch, which is identical to the one from the multimodal FNN architecture. The overall architecture is illustrated in Figure 5. However, similarly to the previous architecture, this architecture also varies slightly between the individual and combined end motif setting. For the individual end motif setting, the architecture follows the structure illustrated in Figure 5. In the combined end motif setting, the architecture handles the jointly processing of the end motifs associated with each fragment using the same approach as the multimodal FNN.

The dimensionality of the input layer from the fragment length branch follows that of the single-feature FNN implemented for fragment length. For the end motif branch, the input consists of the 10×4 matrix representations obtained through one-hot encoding of the end motifs. These representations are processed through the one-dimensional convolutional kernels spanning the nucleotide dimension. The number of kernels, kernel size, and the dimensionality of the hidden layers in the fragment length and feature fusion branch are determined through hyperparameter optimization.

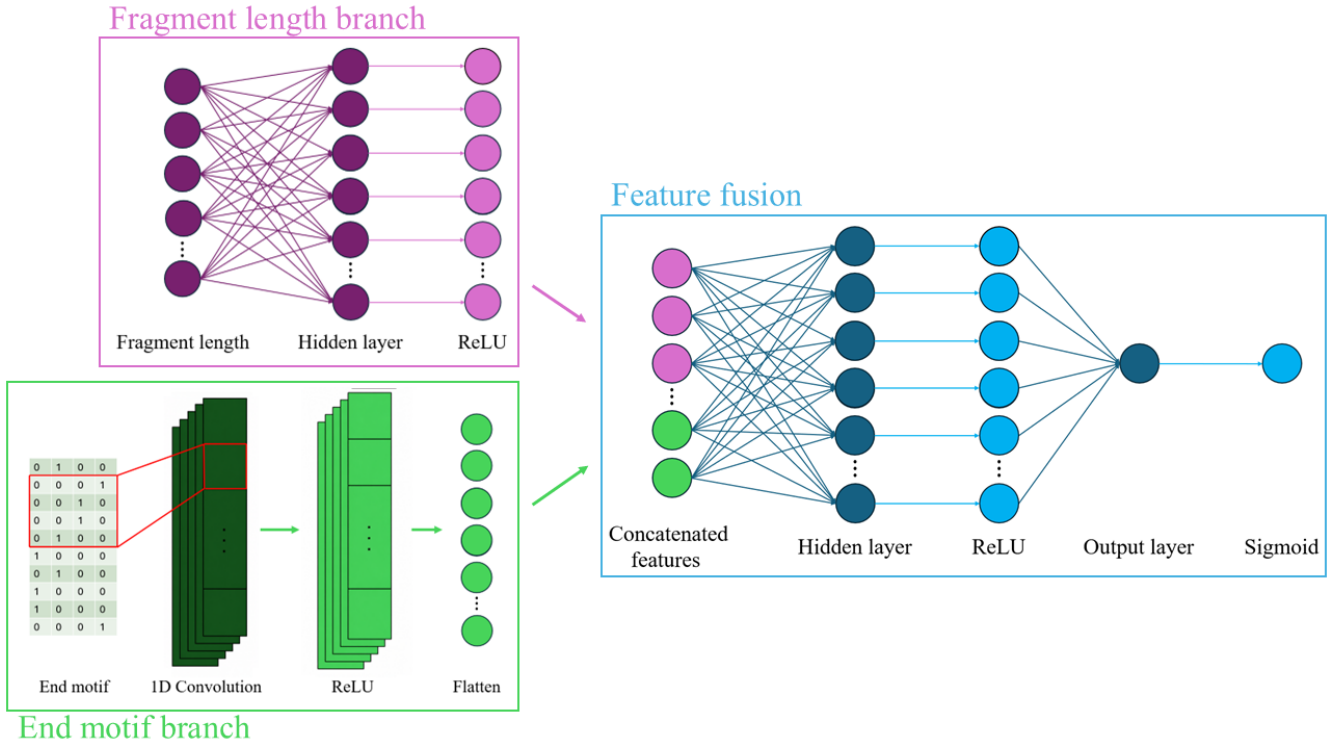


Figure 5: The multimodal CNN architecture.

2.2.3 Recurrent Neural Network

Although the previous section argued that CNN-based architectures may be more suitable than RNN- and Transformer-based architectures for fragment-level end motif modeling, the success these architectures have achieved in genomic sequence-based prediction tasks cannot be ignored [32, 38]. Furthermore, the simpler architecture of RNNs compared to Transformers, combined with their inherent modeling of sequential and positional dependencies, makes them relevant candidates for fragment-level end motif modeling [34, 35]. Since the introduction of the original RNN architectures, the field has undergone extensive development, among which the introduction of Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) architectures have been particularly influential [34, 35]. The LSTM architecture introduced the concept of a gated memory cell, commonly referred to as the cell state, which enables controlled propagation of information across a sequence. This addressed a major limitation of early RNNs, namely the vanishing of sequential information over longer sequences. By regulating the flow of information through gating mechanisms, LSTMs allow important information to be preserved throughout the sequence. The GRU architecture builds upon the concepts introduced by LSTMs by further simplifying the gated memory mechanism [34, 35].

Since their introduction, LSTM and GRU architectures have become some of the most widely used RNN-based models in genomics [32]. In particular, bidirectional variants of these architectures (BI-LSTM and BI-GRU) are commonly applied to DNA sequence modeling, as regulatory and structural genomic signals are often influenced by nucleotide context on both sides of a genomic position [32]. Based on these observations, BI-LSTM and BI-GRU were selected as the basis for the next two model architectures introduced in this thesis. These architectures build on the general structure of the multimodal CNN architecture described in the previous section, by replacing the CNN layer in the end motif branch with a bidirectional recurrent layer.

Bidirectional Long Short-Term Memory

The first of these architectures is using a BI-LSTM layer. As mentioned previously, the model follows the same overall structure as the multimodal CNN architecture, but replaces the CNN layer in the end motif branch with a BI-LSTM layer. Similarly to the multimodal CNN architecture, the input to the end motif branch consists of the 10×4 matrix representations obtained through one-hot encoding of the end motifs. These representations are interpreted as sequences of length 10, where each sequence element is represented by a vector of length 4. The BI-LSTM layer uses two separate LSTM units to process the sequence simultaneously in both a forward and backward direction. Each unit maintains both a hidden state and a cell state throughout the sequence, enabling the propagation and retention of sequential information. The final hidden states from the forward and backward LSTM units are concatenated to form the feature representation of the end motif branch. The remainder of the architecture follows the multimodal CNN architecture. This multimodal BI-LSTM architecture implemented for the individual end motif setting is illustrated in Figure 6. The architecture varies for the combined end motif setting in the same way as the multimodal CNN architecture.

The input dimensionality of the LSTM units is fixed at 4, corresponding to the dimensionality of each sequence element. The dimensionality of the hidden states is determined through hyperparameter

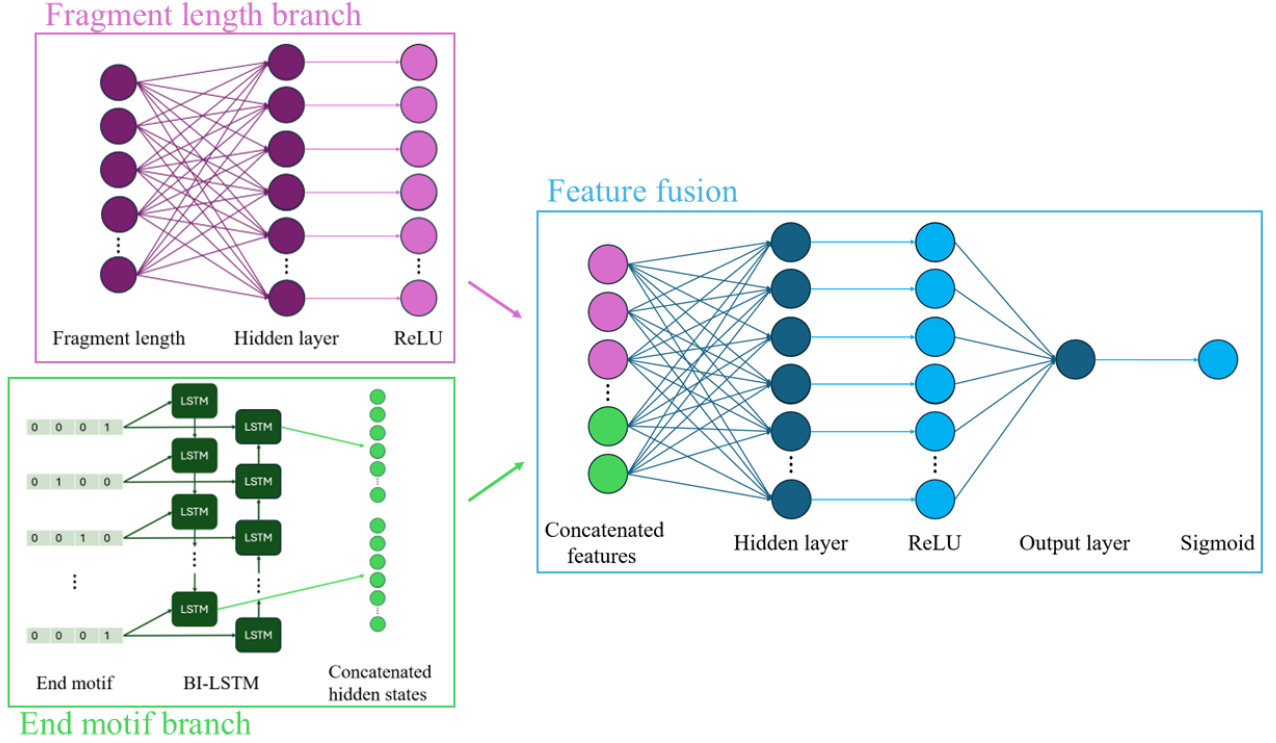


Figure 6: The multimodal BI-LSTM architecture.

optimization. The dimensionality of the hidden layer in the fragment length branch and in the feature fusion branch, respectively, are also determined through hyperparameter optimization.

Bidirectional Gated Recurrent Unit

The second RNN architecture is based on a bidirectional Gated Recurrent Unit (BI-GRU) network. The overall architecture is almost identical to that of the multimodal BI-LSTM architecture described above, with the only difference being that the BI-LSTM layer is replaced by a BI-GRU layer in the end motif branch. The end motif branch still receives the 10×4 one-hot encoded matrix representations of the end motifs as input. These representations are interpreted as sequences of length 10, where each sequence element is represented by a one-dimensional vector of length 4. The BI-GRU layer processes the sequence simultaneously in both a forward and backward direction using two separate GRU units. In contrast to LSTMs, GRUs maintain only a hidden state throughout the sequence and do not employ a separate cell state. The final hidden states from the forward and backward GRU units are concatenated to form the feature representation of the end motif branch. This multimodal BI-GRU architecture implemented for the individual end motif setting is identical to the BI-LSTM architecture shown in Figure 6, but with the LSTM units replaced with GRU units. The architecture varies for the combined end motif setting in the same way as the multimodal BI-LSTM architecture.

The input dimensionality of the GRU units is fixed at 4, corresponding to the dimensionality of each sequence element. The dimensionality of the hidden states is determined through hyperparameter optimization.

2.3 Hyperparameter optimization

The performance of deep learning models is highly dependent on the choice of hyperparameters, including architectural design choices and training configurations. The following section describes how hyperparameter optimization was utilized for each of the main architectures i.e. FNN, CNN, BI-LSTM, and BI-GRU, to select among the possible model variants arising from the different combinations of fragment length preprocessing and end motif setting. Furthermore, it describes how it was used for the FNN architecture to select between the single-feature and multimodal approach. Finally, it explains how hyperparameter optimization was utilized to determine the hyperparameter configuration of the selected model variants.

2.3.1 Initial hyperparameter search

As mentioned, this thesis considers two alternative data representations of the end motifs: the individual versus combined end motif setting. Furthermore, two preprocessing strategies are considered for the fragment length feature: normalization and one-hot encoding. For each of the main architectures, this results in four multimodal model variants implementing fragment length and end motif features. In the case of the FNN architecture, an additional four single-feature variants arise from the single-feature approach: two based exclusively on fragment length and two based exclusively on end motifs. Altogether, this results in a total of 20 distinct model variants. An initial hyperparameter search was carried out to decide on one model variant for each of the main architectures. This entails selecting an end motif setting and fragment length preprocessing for each of the main architectures and further deciding between the single-feature and multimodal approach for the FNN architecture. This initial hyperparameter search was in turn used to generate an exploratory analysis of the hyperparameter configurations’ effect on predictive performance.

Each of the model variants contains architectural components whose configurations are determined through hyperparameter optimization. In addition, hyperparameter optimization was utilized to optimize the performance of the training regime for the model variants. The complete hyperparameter spaces for each of the 20 model variants in this initial hyperparameter search are presented in Table 1. For each of the main architectures, the four multimodal model variants share identical architectural hyperparameter spaces, as the general architecture remains identical between them. The same applies to the single-feature variants of the FNN architecture. Furthermore, the hyperparameter spaces associated with the training regime are identical across all variants. Bayesian optimization was employed to efficiently explore the hyperparameter spaces while minimizing computational cost. The objective for the Bayesian optimization was defined as maximizing the area under the receiver operating characteristic (ROC) curve (AUC) on the validation dataset, described in section 2.1.2. This was done using the Bayesian optimization framework from the AI development platform Weights and Biases. All model configurations were implemented in PyTorch and trained using binary cross entropy (BCE) loss on the training dataset, described in section 2.1.2. During this initial hyperparameter search, each hyperparameter configuration was trained for five epochs. The relatively low number of training epochs was based on the reasoning that hyperparameter configurations with strong performance are also likely to demonstrate favorable learning behavior during the early stages of training[42]. Using a low number of epochs thereby enables evaluation of the model variants and hyperparameter configurations while minimizing computational costs.

(a) Training hyperparameter search spaces (shared across all architectures).

Hyperparameter	Search space
Optimizer	{SGD, RMSProp, Adam}
Learning rate	Log-uniform $[10^{-4}, 10^{-2}]$
Scheduler	{None, Cyclical, Cosine annealing, One-cycle}
Minimum learning rate	Log-uniform $[10^{-5}, 10^{-4}]$
Maximum learning rate	Log-uniform $[10^{-3}, 10^{-1}]$
Momentum	Uniform $[0.80, 0.95]$ (SGD and RMSProp only)
Weight decay	{0, 10^{-5} , 10^{-4} , 10^{-3} , 10^{-2} }
Batch size	{32, 64, 128, 256, 512, 1024}

(b) Architecture-specific hyperparameter search spaces.

Hyperparameter	BI-GRU	BI-LSTM	CNN	FNN
GRU hidden size	{8, 16, 32, 64, 128}	—	—	—
LSTM hidden size	—	{8, 16, 32, 64, 128}	—	—
Number of filters	—	—	{64, 128, 256, 512}	—
Kernel size	—	—	{3, 4, 5}	—
End motif hidden size	—	—	—	{8, 16, 32, 64, 128}
Length hidden size	—	{8, 16, 32, 64, 128}	—	—
Fusion hidden size	—	{64, 128, 256, 512}	—	—

Table 1: Hyperparameter search spaces used during the initial hyperparameter optimization. Architecture-specific hyperparameter search spaces differed between model architectures, whereas the same training-related hyperparameter search space was used for all architectures.

As mentioned, the hyperparameter search was partly employed to optimize the training regime associated with the different model variants. Consequently, multiple optimization and regularization hyperparameters were included in the search space, including optimizer choice, learning rate configuration, learning rate scheduler, momentum, weight decay and batch size. The optimizers included in the search space were Stochastic Gradient Descent (SGD), Root Mean Square Propagation (RMSProp), and the Adaptive Moment Estimation (Adam). These optimizers were selected because they represent some of the most widely used and empirically successful optimization algorithms in deep learning, encompassing both classical stochastic gradient-based optimization and adaptive learning rate methods [43, 44]. Learning rate scheduling was included in the training regime due to previously demonstrated importance of dynamically adapting the learning rate during training of deep neural networks. Specifically, cyclical, cosine annealing, and one-cycle learning rate schedulers were investigated in the hyperparameter optimization based on previous studies reporting improved convergence behavior, reduced training time, and enhanced predictive performance when comparing these schedulers to fixed learning rate strategies [45, 46, 47]. Regularization in the form of weight decay was included as part of the training regime due to its ability to reduce overfitting and improve model generalization. During hyperparameter optimization, weight decay was explored over the grid $[0, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}]$. This search strategy is consistent with that proposed by Leslie N. Smith for the setting where there is no prior assumptions regarding suitable weight decay values

for the corresponding model [42]. Furthermore, small batch sizes have previously been suggested to have a regularizing effect during training. However, for a fixed number of epochs, smaller batch sizes significantly increase computational cost. This means that larger batch sizes may be preferable provided that they do not result in a significant loss of generalization. To identify suitable batch sizes, it was included in the hyperparameter optimization over the grid [32, 64, 128, 256, 512, 1024] exploring a wide range of possible batch sizes [42]. As shown in Table 1, the remainder of the training-related hyperparameters were optimized through continuous hyperparameter sampling from predefined probability distributions. Combining continuous hyperparameter sampling with Bayesian optimization avoids arbitrary discretization of continuous parameter values and enables more fine-grained exploration compared to predefined grid-search strategies. Specifically, the learning rate and the minimum and maximum learning rates associated with the learning rate scheduler were sampled logarithmically from a log-uniform distribution defined over the intervals $[10^{-4}, 10^{-2}]$, $[10^{-5}, 10^{-4}]$, and $[10^{-3}, 10^{-1}]$, respectively. This enables exploration across multiple orders of magnitude, which is important as suitable learning rates can differ by several orders of magnitude between different model architectures and training configurations [48]. Finally, momentum was sampled linearly from a uniform distribution defined over the interval $[0.8, 0.95]$ as momentum values are typically constrained to this relatively narrow interval when implemented for the Root Mean Square Propagation (RMSProp) and Stochastic Gradient Descent (SGD) optimizer. Furthermore, if a cyclical scheduler is chosen, the momentum would also be made cyclical [42]. Momentum is added to the training regime, as it is designed to accelerate network training. However, Adam already inherently incorporates a momentum-like mechanism through exponential moving averages of past gradients, whereas SGD and RMSProp can optionally be extended with momentum-based updates. Consequently, the added momentum in the hyperparameter optimization is only affecting SGD and RMSProp.

All of the architectural hyperparameters are selected for discrete hyperparameter optimization. The grids for these hyperparameters are made to span a wide range of possible complexities, such that the tradeoff between complexity and performance can be investigated.

The mean, maximum and standard deviation of the validation AUC for each model variant was compared in order to facilitate selection among the model variants for each main architecture. This allowed for the selection to be based on a combination of robustness and peak potential performance.

2.3.2 Search space refinement

As described previously, the initial hyperparameter optimization also served as an exploratory analysis of the hyperparameter configurations’ effect on predictive performance. The goal of this analysis was to generate refined search spaces for the selected model variants from the initial hyperparameter search emphasizing high-performing regions. These refined search spaces would then be used to generate an extended hyperparameter search for their corresponding model variant. To investigate the influence of individual hyperparameters on predictive performance, a marginal analysis visualizing the effect of each hyperparameter on the predictive performance was generated. These analyses consisted of plotting the validation AUC versus the values of the individual hyperparameters. These plots provided an initial indication of which hyperparameters appeared most strongly associated with predictive performance. However, interpretation of marginal

relationships alone is insufficient, as the effect of a given hyperparameter is expected to depend on interactions with other hyperparameters. Consequently, an interaction analysis examining the relationship between hyperparameter interactions and validation AUC were also performed. In particular, interactions involving hyperparameters that exhibited strong marginal trends, as well as generally important architectural and training-related interactions, were investigated. These analyses consisted of heatmaps visualizing the mean AUC for specific value combinations of the selected pairs of hyperparameters.

To improve the interpretability of the marginal and interaction analyses, the worst performing 10% of configurations for the model variant ranked according to validation AUC were excluded from the analyses. Bayesian optimization initially explores the search space through partially random sampling before iteratively concentrating sampling toward promising regions based on previously observed performance. Consequently, some of the initial exploratory configurations produced substantially lower validation AUC scores than the remaining configurations, which distorted the visualization of trends within the higher-performing regions of the search space. Removing the worst performing configurations therefore allowed the marginal and interaction plots to more clearly illustrate the characteristics of the high-performing hyperparameter regions most relevant to the refined search spaces.

Based on these analyses, refined hyperparameter search spaces emphasizing high-performing regions of the original search spaces were defined for each of the model variants to facilitate the extended hyperparameter optimization.

2.3.3 Extended hyperparameter search

An extended hyperparameter optimization was subsequently conducted for the model variants selected during the initial hyperparameter search using their respective refined search spaces. The primary objective of this extended optimization phase was to allow each hyperparameter configuration to complete training and thereby enable a more reliable comparison of configuration performance. Consequently, the maximum number of training epochs was increased to 50. To reduce the risk of overfitting and avoid unnecessary training, early stopping with a patience of 10 epochs was employed. For each hyperparameter configuration, model performance was evaluated using the model state corresponding to the epoch achieving the lowest validation loss. As with the initial hyperparameter search, Bayesian optimization with the goal of maximizing validation AUC was used to effectively explore the search space. Furthermore, all model configurations were implemented in PyTorch and trained using BCE loss on the training dataset.

Subsequently, the ten configurations achieving the highest validation AUC scores for each architecture were selected for further evaluation. To assess the robustness and consistency of the selected configurations, each of the ten configurations for a given architecture was re-trained ten times using different random seeds. The re-training followed the same procedure as training in the extended hyperparameter search. Final hyperparameter selection for each model variant was then based on comparison of the mean, maximum, and standard deviation of the AUC values obtained across these re-training runs. Finally, the selected hyperparameter configuration for each model variant was re-trained on the combined training and validation datasets. Including the validation dataset in the training removes the possibility of utilizing early stopping. Therefore,

the number of training epochs used during this final training phase was determined based on an analysis of the training and validation loss curves obtained during the re-training experiments for the corresponding configuration.

2.4 Model evaluation

To compare the predictive performance of the four main architectures, the final model variants obtained from the extended hyperparameter optimization were evaluated on the independent test dataset. As described in Section 2.1.2, the test dataset contains fragment-level labels, enabling direct evaluation of fragment-level classification performance. An initial comparison of the architectures was therefore performed by calculating the AUC for fragment-level predictions on the test dataset. This evaluation provided insight into the ability of the architectures to distinguish ctDNA fragments from non-tumor-derived cfDNA fragments at the level of individual fragments. To investigate whether the proposed deep learning architectures provided improvements over more conventional machine learning approaches, a fragment-level logistic regression baseline model implementing both fragment length and end motif features was included in the evaluation. The logistic regression model was implemented using the `LogisticRegression` class from the `scikit-learn` Python library. Similar to the deep learning architectures, the logistic regression model was trained using the combined training and validation datasets and subsequently evaluated on the independent test dataset.

As mentioned in the introduction, a clinically relevant implementation of the models would ultimately require ctDNA detection at sample-level. Therefore, an additional evaluation framework was developed to investigate whether the fragment-level predictions generated by the models could support accurate sample-level ctDNA classification. Since the implemented architectures produce predictions for individual fragments only, this required both construction of a sample-level test dataset and definition of a strategy for aggregating fragment-level predictions into a single sample-level prediction.

To facilitate this clinically relevant evaluation, a simulated sample-level dataset was generated from the fragment-level test dataset. Each sample in this simulated dataset consisted of 100,000 randomly selected fragments from the fragment-level test dataset. This sampling was not done completely random, as each sample was made to have a fixed fraction of ctDNA fragments. Using this procedure, 100 simulated samples were generated for each of the following ctDNA fractions: 0.0, 0.001, 0.002, 0.004, 0.006, 0.008, 0.01, and 0.02. This simulation framework enabled controlled investigation of model performance across varying ctDNA concentrations.

Several aggregation strategies were subsequently evaluated for converting fragment-level predictions into sample-level predictions. Specifically, the mean prediction score and the fraction of fragment predictions exceeding thresholds of 0.5, 0.6, 0.7, 0.8, and 0.9 were investigated. To determine the most effective aggregation strategy for each architecture, ROC curves and corresponding AUC values were generated for classification between simulated samples with ctDNA fractions of 0.0 and 0.004. The ctDNA fraction of 0.004 was selected because it represents a relatively low but clinically relevant ctDNA level, thereby approximating a realistic early-detection setting. The best-performing aggregation strategy identified for each architecture was subsequently used to repeat the classification analysis across the remaining ctDNA fractions. This enabled evaluation of the

sensitivity of each architecture for decreasing ctDNA fractions and thereby facilitated identification of the ctDNA levels at which reliable sample-level classification remained achievable.

2.5 SHAP analysis

Deep learning models are often criticized for their limited interpretability, as their predictions are typically generated through highly complex and non-linear transformations of the input features. The deep learning models introduced in this thesis are no exception. None of the model architectures allow for a clear interpretation of the effect of the different input features on the predictions. As previously mentioned, the biological mechanics governing the cfDNA concentration and fragmentomics remain poorly understood. It would therefore be highly beneficial if it was possible to infer how the different fragmentomic features contributed to the model predictions.

To facilitate such an investigation of the final models, a SHapley Additive exPlanations (SHAP) analysis was performed. SHAP is a model explainability framework based on Shapley values from cooperative game theory. In 1953, Lloyd Shapley defined a set of properties that a fair measurement of contribution in a collaborative game should fulfill. Lloyd Shapley further proved that the only measurement which satisfies all these properties are the Shapley values [49]. For a set of players $N = \{1, 2, \dots, n\}$ the Shapley value for player $j \in N$ is defined as:

$$\phi_j = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} (V(S \cup \{j\}) - V(S)) \quad (1)$$

where S is a coalition (i.e. subset) of players, $V(S)$ is the payout of the game for the coalition S and $(V(S \cup \{j\}) - V(S))$ is a quantitative measure of the marginal contribution of player j to coalition S . The Shapley values can thereby be interpreted as the weighted average of a player’s marginal contributions across all possible coalitions. The use of Shapley values for ML interpretability was made popular by Lundberg et al. [50] who introduced the SHAP framework. With the SHAP framework, Lundberg et al. introduced SHAP values which can be used to measure the influence of the model features on the model predictions. The SHAP values are a translation of the Shapley values from the field of game theory to the field of ML. This translation is made by interpreting each player as a feature, the payout as the predicted value of the model and each observation as a new cooperative game [50]. SHAP values are therefore also calculated locally for each observation rather than one global value per feature.

Since the SHAP values simply are a translation of the Shapley values, they are also derived from the original Shapley value formulation (1). Consequently, exact SHAP values satisfy the same properties as defined by Lloyd Shapley for the Shapley values. Among these, the property of efficiency provides a definition that is crucial for the interpretation of the SHAP values. Consider a setting where a model f is defined for a set of features N , and x_i is an observation from this feature space. Then the SHAP values for x_i satisfy the following definition:

$$f(x_i) = \phi_0 + \sum_{j \in N} \phi_{i,j}$$

where ϕ_0 is the average prediction for the model and $\phi_{i,j}$ is the SHAP value of the j ’th feature for x_i . Based on this definition, the SHAP values assigned to each feature $j \in N$ for observation x_i defines

to which degree the feature j contributed to the difference between the individual prediction of x_i and the average model prediction. The SHAP values thereby provide a measure of how the different input features contribute to the model predictions. However, calculating exact SHAP values using the original Shapley value formulation (1) is computationally infeasible for deep learning models, and therefore a multitude of approximation techniques have been introduced [51]. In this thesis, the `GradientExplainer` class from the `shap` Python library was used to generate SHAP value approximations for the final models. This class uses expected gradients to approximate the SHAP values. Expected gradients is an extension of the integrated gradients method introduced by Sundararajan et al. [52]. The method is designed for differentiable models and approximates SHAP values by estimating how changes in the input features influence the model predictions relative to a background distribution of observations [53]. The `GradientExplainer` class was chosen due to its applicability to arbitrary differentiable models and its straightforward implementation for models defined in Pytorch.

In the context of this thesis, the SHAP analysis was used to estimate how individual fragmentomic features, such as fragment lengths and nucleotide positions within end motifs, contributed to the predicted probability of a fragment being ctDNA. To obtain global measures of feature importance, the mean absolute SHAP value was calculated across observations for each feature.

3 Results

In the following section, the results obtained through the selection, training and evaluation of the different architectures will be presented. The results were obtained following the approach described in section 2.

3.1 Hyperparameter optimization

3.1.1 Initial hyperparameter search

As described in Section 2.3.1, the primary objective of the initial hyperparameter search was to determine a single model variant for each of the main architectures. To support such a selection, Table 2 was generated, summarizing the mean, maximum, and standard deviation of the AUC values obtained during the initial hyperparameter optimization for the 20 model variants.

When comparing the single-feature and multimodal FNN variants, the results presented in Table 2 indicate that the multimodal variants consistently achieved the strongest predictive performance. Across all combinations of fragment length preprocessing and end motif setting, the multimodal variants obtained higher mean and maximum AUC values than any of the corresponding single-feature variants. Furthermore, the single-feature variants based exclusively on fragment length achieved substantially stronger performance than the end motif-only variants, suggesting that fragment length constitutes the dominant predictive signal. Nevertheless, incorporating end motif information alongside fragment length further improved predictive performance, indicating that end motifs provide complementary information despite their limited standalone predictive power, thereby supporting the multimodal approach. Finally, the multimodal and fragment length-only

BI-GRU					CNN				
Length preprocessing	End motif setting	mean	max	sd	Length preprocessing	End motif setting	mean	max	sd
one-hot	individual	0.6814	0.6938	0.0399	one-hot	individual	0.6833	0.6935	0.0327
one-hot	combined	0.7010	0.7059	0.0161	one-hot	combined	0.6935	0.7037	0.0320
normalize	individual	0.6906	0.6951	0.0091	normalize	individual	0.6878	0.6946	0.0205
normalize	combined	0.6975	0.7071	0.0229	normalize	combined	0.6919	0.7054	0.0314

BI-LSTM					FNN				
Length preprocessing	End motif setting	mean	max	sd	Length preprocessing	End motif setting	mean	max	sd
one-hot	individual	0.6800	0.6938	0.0423	one-hot	individual	0.6861	0.6930	0.0254
one-hot	combined	0.6972	0.7059	0.0283	one-hot	combined	0.6993	0.7042	0.0182
normalize	individual	0.6888	0.6955	0.0166	normalize	individual	0.6899	0.6950	0.0088
normalize	combined	0.7014	0.7071	0.0139	normalize	combined	0.6999	0.7060	0.0137

FNN - Length only					FNN - End motifs only				
Length preprocessing	mean	max	sd		End motif setting	mean	max	sd	
normalize	0.6724	0.6786	0.0102		individual	0.5802	0.5873	0.0095	
one-hot	0.6703	0.6755	0.0207		combined	0.6069	0.6120	0.0089	

Table 2: Summary statistics of validation AUC from the initial hyperparameter search for each of the 20 model variants.

variants exhibited comparable standard deviations, suggesting similar levels of robustness between these approaches. Based on these observations, the multimodal model approach was selected for the FNN architecture.

Table 2 further reveals a consistent pattern regarding the selection of end motif setting. Across all architectures, the model variants implementing the combined end motif setting achieved both higher mean and maximum AUC values than their counterparts using the individual end motif setting, regardless of the fragment length preprocessing strategy. This indicates that the combined end motif representation consistently provided stronger predictive performance than the individual representation. However, although this pattern was observed consistently across architectures, the absolute performance differences between the two end motif settings were relatively small. Nevertheless, the observed trend resulted in the combined end motif setting being selected for all the main architectures.

In contrast, no equally consistent pattern was observed regarding fragment length preprocessing. Although the number of model variants had been substantially reduced following exclusion of the single-feature approaches and the individual end motif setting, neither normalization nor one-hot encoding consistently outperformed the alternative across all architectures. The only consistent observation was that normalization achieved the highest maximum AUC value for each architecture. However, the overall differences in predictive performance between the preprocessing strategies were minimal, making it difficult to identify a clearly superior preprocessing approach for the individual architectures. Since the preprocessing strategy appeared to have limited influence on predictive performance, the final selection was based primarily on computational simplicity. Normalization was selected as the fragment length preprocessing strategy for each of the main architectures, as it substantially reduces the input dimensionality and number of trainable parameters compared to

one-hot encoding, thereby resulting in a computationally simpler model architecture.

The initial hyperparameter optimization thereby resulted in the multimodal approach being selected for the FNN architecture. Furthermore, it resulted in the model variant implementing normalization and the combined end motif setting being selected for each of the main architectures.

3.1.2 Search space refinement

As mentioned in section 2.3.2, the initial hyperparameter search was further utilized to investigate the relationship between hyperparameter configurations and predictive performance in order to generate refined search spaces for the extended hyperparameter search. These refined search spaces would emphasize high-performing regions of the original search spaces and define personalized training hyperparameter search spaces for each architecture. As mentioned, this refinement was facilitated through a series of plots analyzing both the marginal effect and the interaction effect of the hyperparameters for each main architecture.

The following sections will present the results from the marginal and interaction analyses for each of the main architectures. It should be noted that the observed marginal and interaction effects on predictive performance are generally modest, with most differences in validation AUC occurring

(a) Training hyperparameter search spaces.

Hyperparameter	BI-GRU	BI-LSTM
Optimizer	{Adam}	{Adam}
Scheduler	{Cyclical, Cosine, One-cycle}	{None, Cyclical, One-cycle}
Learning rate	$[10^{-4}, 10^{-2}]$	$[10^{-4}, 10^{-2}]$
Weight decay	$\{10^{-4}, 10^{-3}, 10^{-2}\}$	$\{0, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}\}$
Batch size	{128, 256, 512, 1024}	{64, 128, 256, 512}

Hyperparameter	CNN	FNN
Optimizer	{Adam, SGD}	{Adam}
Scheduler	{None, Cosine, One-cycle}	{None, Cyclical, Cosine, One-cycle}
Learning rate	$[10^{-4}, 4 \times 10^{-3}]$	$[3 \times 10^{-4}, 4 \times 10^{-3}]$
Weight decay	$\{10^{-5}, 10^{-4}, 10^{-3}\}$	$\{0, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}\}$
Batch size	{128, 256, 512, 1024}	{128, 256, 512}

(b) Architecture-specific hyperparameter search spaces.

Hyperparameter	BI-GRU	BI-LSTM	CNN	FNN
GRU hidden size	{32, 64, 128, 256}	—	—	—
LSTM hidden size	—	{32, 64, 128, 256}	—	—
Number of filters	—	—	{64, 128, 256}	—
Kernel size	—	—	{4, 5, 6}	—
End motif hidden size	—	—	—	{32, 64, 128, 256}
Length hidden size	—	{32, 64, 128, 256}	—	—
Fusion hidden size	—	{64, 128, 256, 512}	—	—

Table 3: Refined hyperparameter search spaces used during the extended hyperparameter search. The search spaces were refined based on the results obtained from the initial hyperparameter search. Training hyperparameter search spaces differed between model architectures in the refined search spaces. The original search space for momentum is still applied for CNN when implementing SGD.

at the third decimal level. This suggests that the predictive performance of the investigated architectures is not strongly sensitive to individual hyperparameter configurations within the explored search spaces. However, the AUC scores presented in the analyses below are based on fragment-level predictions. Since the final sample-level prediction is obtained through aggregation of a large number of fragment-level predictions, even small improvements in fragment-level predictive performance may translate into more substantial improvements at the sample-level. The refined search space for each of the architectures created based on the analyses below is illustrated in Table 3.

FNN

The marginal effects of the hyperparameters for the FNN architecture implementing normalization and the combined end motif setting are illustrated in Figure 7. The figure indicates that several hyperparameters exhibit clear associations with predictive performance. The strongest effects were observed for the choice of optimizer and the dimensionality of the hidden layer in the end motif branch. In particular, the Adam optimizer consistently outperformed both RMSProp and SGD, while increasing the dimensionality of the hidden layer in the end motif branch was associated with improved validation AUC. A similar, although less pronounced, trend was observed for the dimensionality of the hidden layer in the fragment length branch, where larger hidden dimensions likewise appeared to improve predictive performance. The learning rate also demonstrated a moderate marginal effect, with intermediate values within the explored range generally outperforming values near the boundaries of the search space. A similar pattern was observed for the batch size hyperparameter.

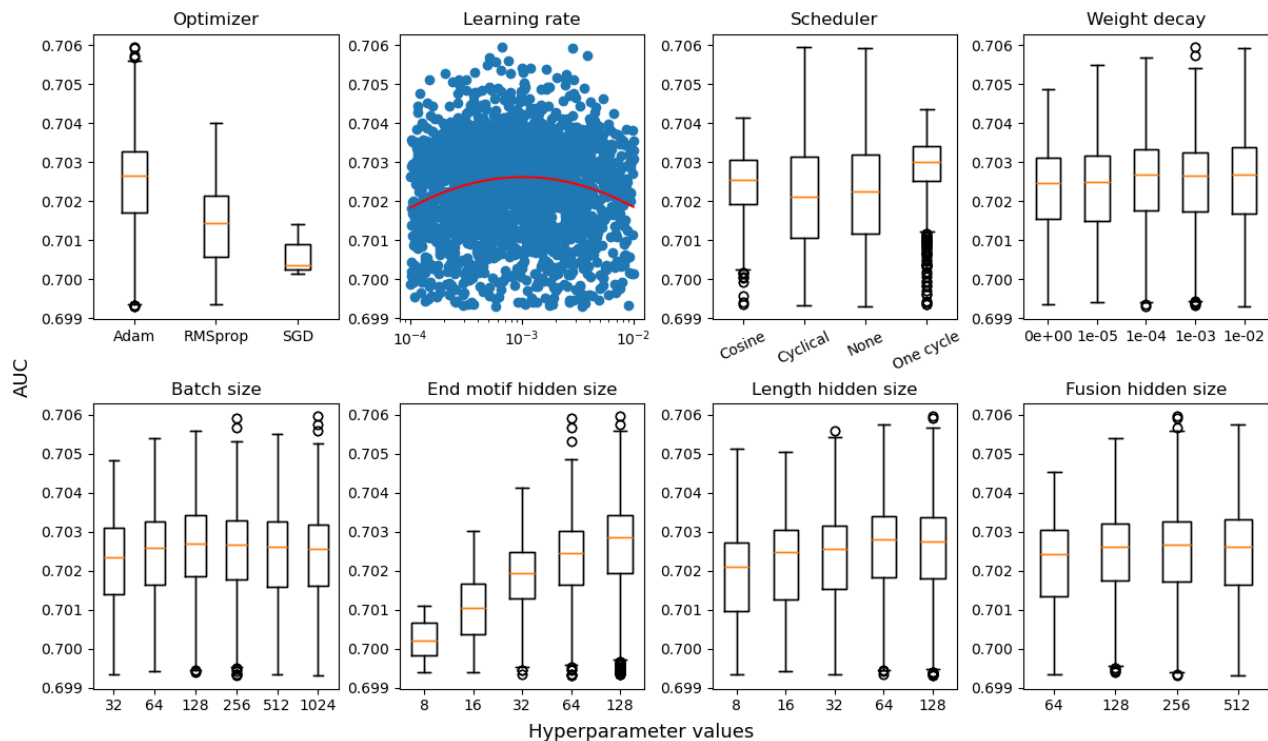


Figure 7: The marginal effects of the different hyperparameters on validation AUC for the FNN architecture. The effect of the continuous learning rate is shown by fitting a second degree polynomial.

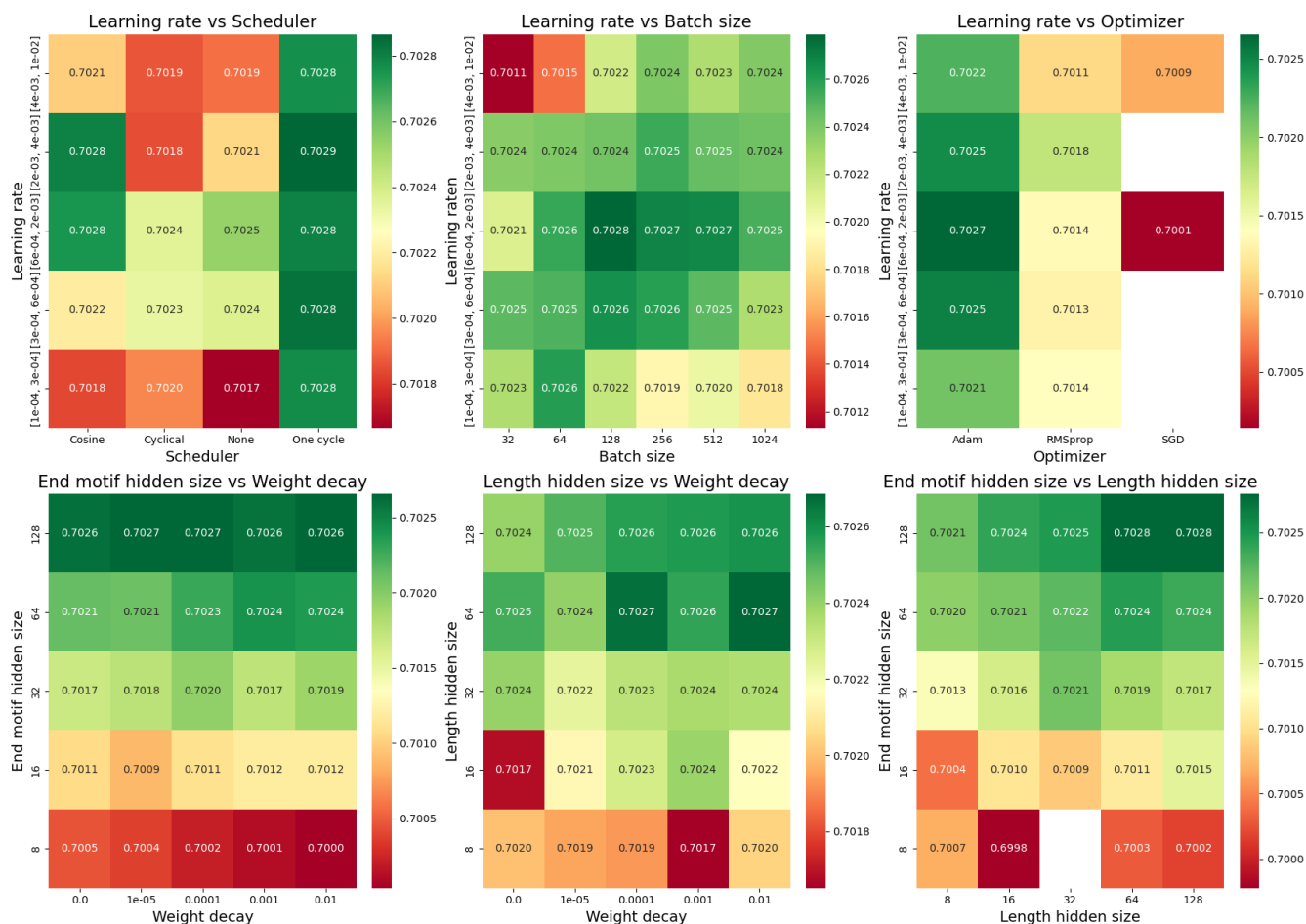


Figure 8: The effect of different hyperparameter interactions on validation AUC for the FNN architecture. For each interaction the mean validation AUC for all configurations implementing the corresponding interaction is shown.

The heatmaps illustrating the selected hyperparameter interactions for the FNN architecture are shown in Figure 8. Interactions involving the hidden layer dimensionalities of both the fragment length branch and the end motif branch were included, as these hyperparameters demonstrated strong marginal effects on predictive performance. In particular, the interactions between the hidden layer dimensionalities and weight decay were investigated to assess whether larger model capacities benefited from stronger regularization. Furthermore, the interaction between the hidden layer dimensionalities of the two branches was examined to investigate whether simultaneously increasing the capacity of both branches improved predictive performance. The interactions between learning rate and scheduler, learning rate and optimizer, and learning rate and batch size were additionally investigated due to the expected coupling between these training-related hyperparameters.

The interaction analysis generally supported the trends observed in the marginal analysis. In particular, Adam consistently outperformed both RMSProp and SGD across the explored learning rate regions. Furthermore, increasing hidden layer dimensionality generally improved predictive performance regardless of the applied weight decay. The interaction analysis additionally supported the preference for moderate learning rates and batch sizes observed in the marginal analysis. These

observations motivated a refined search space emphasizing Adam-based optimization, larger hidden dimensions, and moderate learning rates and batch sizes for the extended hyperparameter search of the FNN architecture.

CNN

The marginal effects of the hyperparameters for the CNN architecture implementing normalization and the combined end motif setting are illustrated in Figure 9. As observed for the FNN architecture, several hyperparameters exhibited clear associations with predictive performance. The strongest effects were observed for the choice of optimizer, kernel size, and the dimensionality of the hidden layer in the fragment length branch. The trends observed for optimizer choice and hidden layer dimensionality were similar to those identified for the FNN architecture, with Adam and larger hidden dimensions generally associated with improved predictive performance. Furthermore, increasing kernel size appeared to improve validation AUC, suggesting that relatively larger local motifs are important for ctDNA prediction. More subtle trends were observed for learning rate and batch size, where smaller learning rates and larger batch sizes generally appear favorable. Finally, the cyclical scheduler generally demonstrated the weakest performance.

As for the FNN architecture, interaction heatmaps were generated for hyperparameters exhibiting

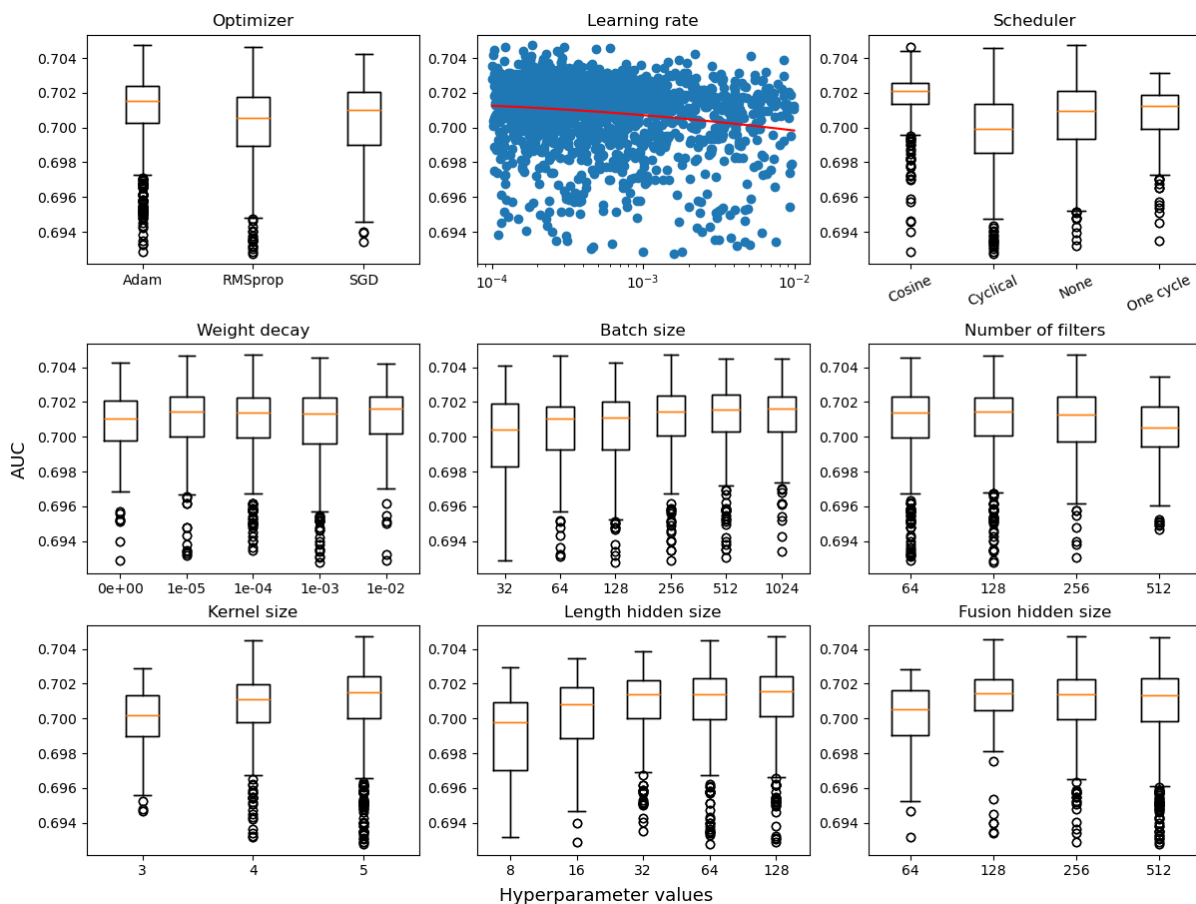


Figure 9: The marginal effects of the different hyperparameters on validation AUC for the CNN architecture. The effect of the continuous learning rate is shown by fitting a second degree polynomial.

strong marginal effects together with training-related hyperparameters expected to demonstrate strong coupling. The interaction analysis primarily supported the trends identified in the marginal analysis and did not reveal substantial additional effects. The corresponding interaction heatmaps for the CNN architecture are included in Appendix A. Based on these observations, a refined search space emphasizing Adam-based optimization, larger hidden dimensions, larger kernel sizes, larger batch sizes, smaller learning rates, and exclusion of the cyclical scheduler was defined for the extended hyperparameter search of the CNN architecture.

BI-LSTM

The marginal effects of the hyperparameters for the BI-LSTM architecture implementing normalization and the combined end motif setting are illustrated in Figure 10. Compared to the CNN and FNN architectures, stronger marginal effects were observed for the BI-LSTM architecture. In particular, the hidden-state dimensionality of the LSTM unit and the dimensionality of the hidden layer in the fragment length branch appeared to strongly influence predictive performance, with larger dimensionalities generally associated with improved validation AUC. Furthermore, none of the configurations utilizing the SGD optimizer were included among the top-performing 90% of configurations, while Adam generally outperformed RMSProp within this subset. Finally, the cosine annealing scheduler consistently underperformed compared to the remaining learning rate schedulers.

As with the previous architectures, interaction heatmaps were generated for hyperparameters exhibiting strong marginal effects together with training-related hyperparameters expected to

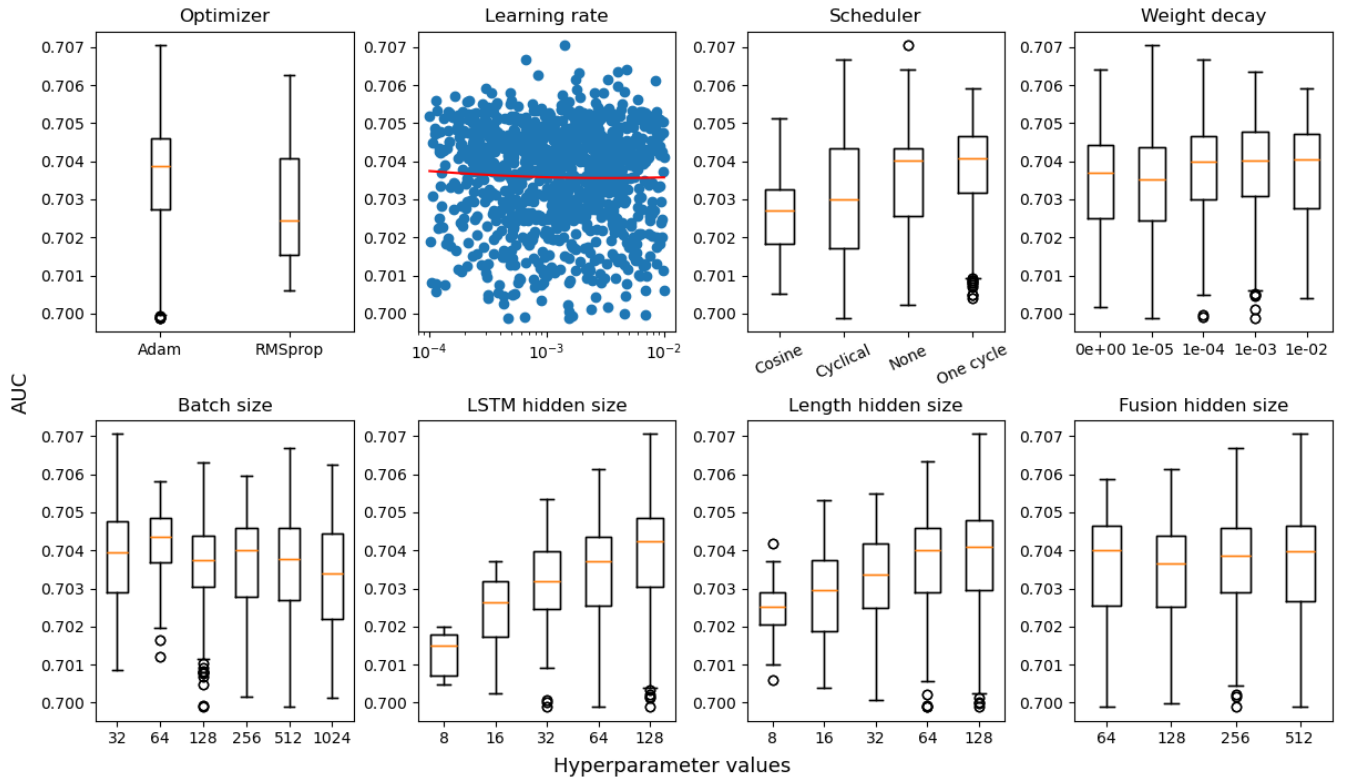


Figure 10: The marginal effects of the different hyperparameters on validation AUC for the BI-LSTM architecture. The effect of the continuous learning rate is showed by fitting a second degree polynomial.

demonstrate strong coupling. The interaction analysis did not substantially alter the conclusions drawn from the marginal analysis. The corresponding interaction heatmaps for the BI-LSTM architecture are included in Appendix B. Given the above observation a refined search space emphasizing Adam-based optimization, larger dimensionalities of hidden state and hidden layer, and exclusion of the cosine annealing scheduler was defined for the extended hyperparameter search of the BI-LSTM architecture.

BI-GRU

The marginal effects of the hyperparameters for the BI-GRU architecture implementing normalization and the combined end motif setting are illustrated in Figure 11. Similar to the previous architectures, the marginal analysis were used to infer relationships between the hyperparameters and predictive performance for the BI-GRU architecture. Furthermore, as for the previous architectures, the interaction analysis did not reveal substantial additional information beyond the trends already observable from the marginal plots. The corresponding interaction heatmaps for the BI-GRU architecture are included in Appendix C. Compared to the previous architectures, the BI-GRU architecture appeared more sensitive to regularization-related hyperparameters and generally preferred higher weight decay values. Based on these observations, a refined search space emphasizing Adam-based optimization, larger hidden-state dimensionalities in the GRU units, larger hidden-layer dimensionalities in the fragment length branch, larger weight decay values and batch sizes, together with mandatory implementation of a learning rate scheduler, was defined for the extended hyperparameter search.

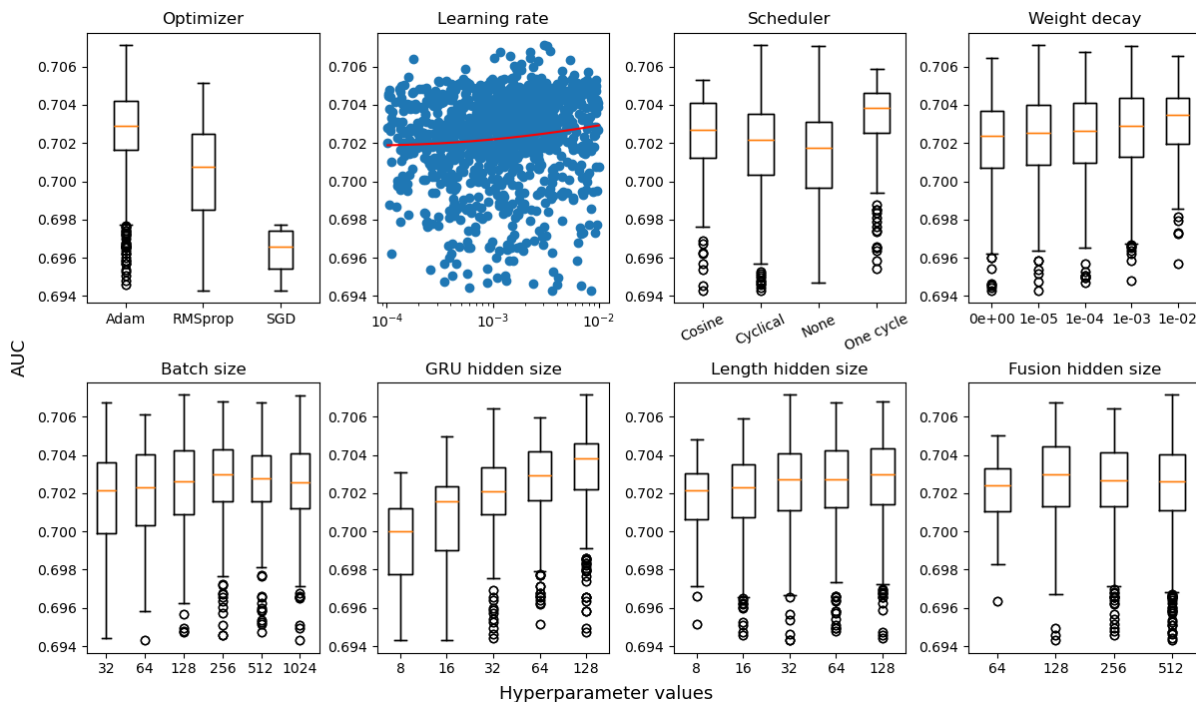


Figure 11: The marginal effects of the different hyperparameters on validation AUC for the BI-GRU architecture. The effect of the continuous learning rate is showed by fitting a second degree polynomial.

Rank	FNN			CNN			BI-LSTM			BI-GRU		
	Mean	Max	SD	Mean	Max	SD	Mean	Max	SD	Mean	Max	SD
1	0.7020	0.7048	0.0025	0.6997	0.7027	0.0018	0.7019	0.7039	0.0028	0.7035	0.7058	0.0021
2	0.7028	0.7047	0.0017	0.7024	0.7035	0.0008	0.7034	0.7062	0.0022	0.7029	0.7055	0.0018
3	0.7021	0.7053	0.0019	0.7013	0.7024	0.0012	0.7034	0.7074	0.0021	0.7024	0.7060	0.0029
4	0.7025	0.7045	0.0014	0.7024	0.7059	0.0025	0.7018	0.7034	0.0016	0.7031	0.7068	0.0024
5	0.7022	0.7045	0.0020	0.7023	0.7046	0.0016	0.7038	0.7060	0.0018	0.7028	0.7058	0.0027
6	0.7028	0.7056	0.0016	0.7027	0.7043	0.0015	0.7038	0.7068	0.0019	0.7026	0.7054	0.0023
7	0.7028	0.7051	0.0014	0.7022	0.7045	0.0015	0.7019	0.7068	0.0049	0.7016	0.7070	0.0033
8	0.7025	0.7053	0.0014	0.7020	0.7039	0.0017	0.7040	0.7073	0.0019	0.7028	0.7074	0.0039
9	0.7012	0.7050	0.0025	0.7018	0.7047	0.0014	0.7026	0.7061	0.0031	0.7042	0.7055	0.0012
10	0.7029	0.7063	0.0017	0.7011	0.7040	0.0026	0.7029	0.7047	0.0017	0.7022	0.7055	0.0021

Table 4: Summary statistics of validation AUC from re-training the top 10 hyperparameter configurations for the selected model variant from each architecture. The hyperparameter configuration selected for each architecture based on these results is shown in bold.

3.1.3 Extended hyperparameter search

Using the refined search spaces illustrated in Table 3, the extended hyperparameter optimization described in Section 2.3.3 was conducted for the selected model variant from each architecture. Subsequently, the ten highest-performing hyperparameter configurations for each selected model variant, ranked according to validation AUC, were identified and re-trained using different seeds. The mean, maximum, and standard deviation of the validation AUC values obtained across these re-training runs are summarized in Table 4. Based on these results, a final hyperparameter configuration was selected for each selected model variant, indicated in bold in Table 4. The selection was based on a compromise between high mean and maximum validation AUC values (reflecting predictive performance) and low standard deviation (reflecting robustness across re-training runs). The selected hyperparameter configurations are shown in Table 5.

To determine the number of epochs used during the final training phase for the selected model variants, the training and validation loss curves from the re-training runs of the selected hyper-

Hyperparameter	BI-GRU	BI-LSTM	CNN	FNN
Optimizer	Adam	Adam	Adam	Adam
Learning rate	1.7×10^{-4}	1.6×10^{-3}	3.6×10^{-4}	5.7×10^{-4}
Scheduler	One cycle	None	Cosine annealing	Cosine annealing
Weight decay	10^{-4}	10^{-2}	10^{-5}	10^{-3}
Batch size	512	128	256	512
GRU hidden size	128	—	—	—
LSTM hidden size	—	128	—	—
Number of filters	—	—	256	—
Kernel size	—	—	6	—
End motif hidden size	—	—	—	64
Length hidden size	256	64	128	256
Fusion hidden size	128	512	512	256

Table 5: Final hyperparameter configurations for the selected model variant from each architecture.

parameter configurations were analyzed visually. This analysis is shown in Figure 12, which illustrates the training and validation loss trajectories for each re-training run for the four selected model variants. The figure additionally indicates the epoch corresponding to the model state with minimum validation loss for each run. Across all architectures, the validation loss generally showed limited improvement beyond approximately 12 training epochs. Only the FNN architecture exhibited a small number of re-training runs in which the epoch with the minimum validation loss exceeded 12. Moreover, for the BI-LSTM and CNN architectures, the highest epoch associated with minimum validation loss across all re-training runs was only 8 and 5, respectively. Finally, all of the architectures exhibited increasing divergence between training and validation loss within the 10 epoch early stopping patience window, suggesting early signs of overfitting during training. These observations combined indicate that extending training beyond the current limit of 50 or increasing the early stopping patience would likely not be beneficial.

Despite variability between re-training runs, architecture-specific patterns regarding the optimal number of training epochs were observed. For the BI-GRU architecture, the minimum validation loss was most frequently achieved between epochs 6 and 12. Consequently, the final training

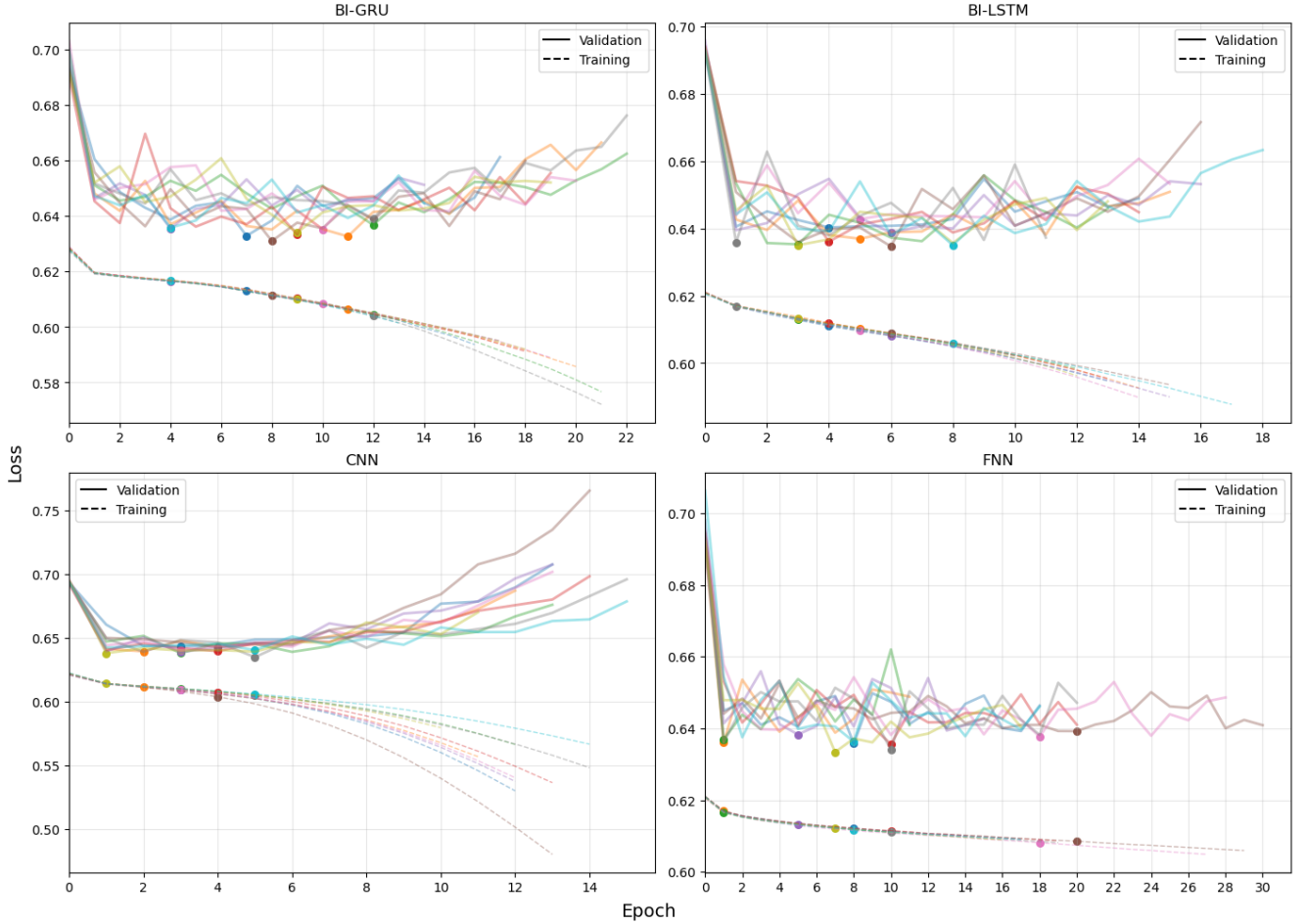


Figure 12: The training and validation loss from the 10 re-training runs of the selected hyperparameter configuration. The mark on each run indicates the epoch corresponding to the model state with the lowest validation loss for the training run.

duration for the BI-GRU architecture was set to 9 epochs. Similarly, the BI-LSTM architecture most commonly achieved minimum validation loss between epochs 3 and 6, resulting in a final training duration of 5 epochs. For the CNN architecture, all epochs corresponding to minimum validation loss were observed within the interval from 1 to 6 epochs with several runs clustering around 3 epochs. Consequently, the final training duration for the CNN architecture was set to 3 epochs. In contrast, the FNN architecture exhibited substantially greater variability in the epochs corresponding to minimum validation loss, ranging from 1 to 20 epochs across re-training runs. This variability seemed to be caused by a fluctuating validation loss, which after the initial training epochs fluctuates around a relatively stable level without displaying a clear trend toward either improvement or deterioration. Nevertheless, a moderate concentration of runs achieved minimum validation loss between epochs 5 and 10. Based on these observations, the final training duration for the FNN architecture was set to 8 epochs.

3.2 Model evaluation

An initial evaluation of the fragment-level predictive performance of the selected model variant from each architecture was performed on the test dataset. The corresponding AUC values for each architecture are shown in Table 6.

Architecture	Test AUC	Test Loss
FNN	0.6888	0.6482
CNN	0.6871	0.6484
BI-LSTM	0.6894	0.6474
BI-GRU	0.6849	0.6495
Log. Reg.	0.6150	0.6520

Table 6: Fragment-level test AUC and loss for the selected model variant from each architecture trained on the combined training and validation dataset.

As could be anticipated, all architectures achieved lower AUC values on the independent test dataset than on the validation dataset during hyperparameter optimization. This likely reflects mild implicit optimization toward validation performance during hyperparameter selection. Furthermore, all deep learning architectures outperformed the logistic regression baseline, indicating that deep learning approaches provide improved fragment-level predictive performance compared with conventional linear machine learning. Among the deep learning architectures, the BI-LSTM achieved the highest fragment-level test AUC. However, the overall performance differences between the deep learning architectures were minimal, with only 0.0045 separating the highest- and lowest-performing architectures. This limited performance difference suggests that the increased architectural complexity associated with more advanced end motif modeling may provide only modest improvements in fragment-level predictive performance. This interpretation is further supported by the strong performance of the comparatively simple FNN architecture, which achieved a test AUC of 0.6888, only 0.0006 below that of the BI-LSTM architecture. Nevertheless, as discussed previously, relatively small improvements at the fragment-level may still translate into more substantial performance differences in the sample-level setting.

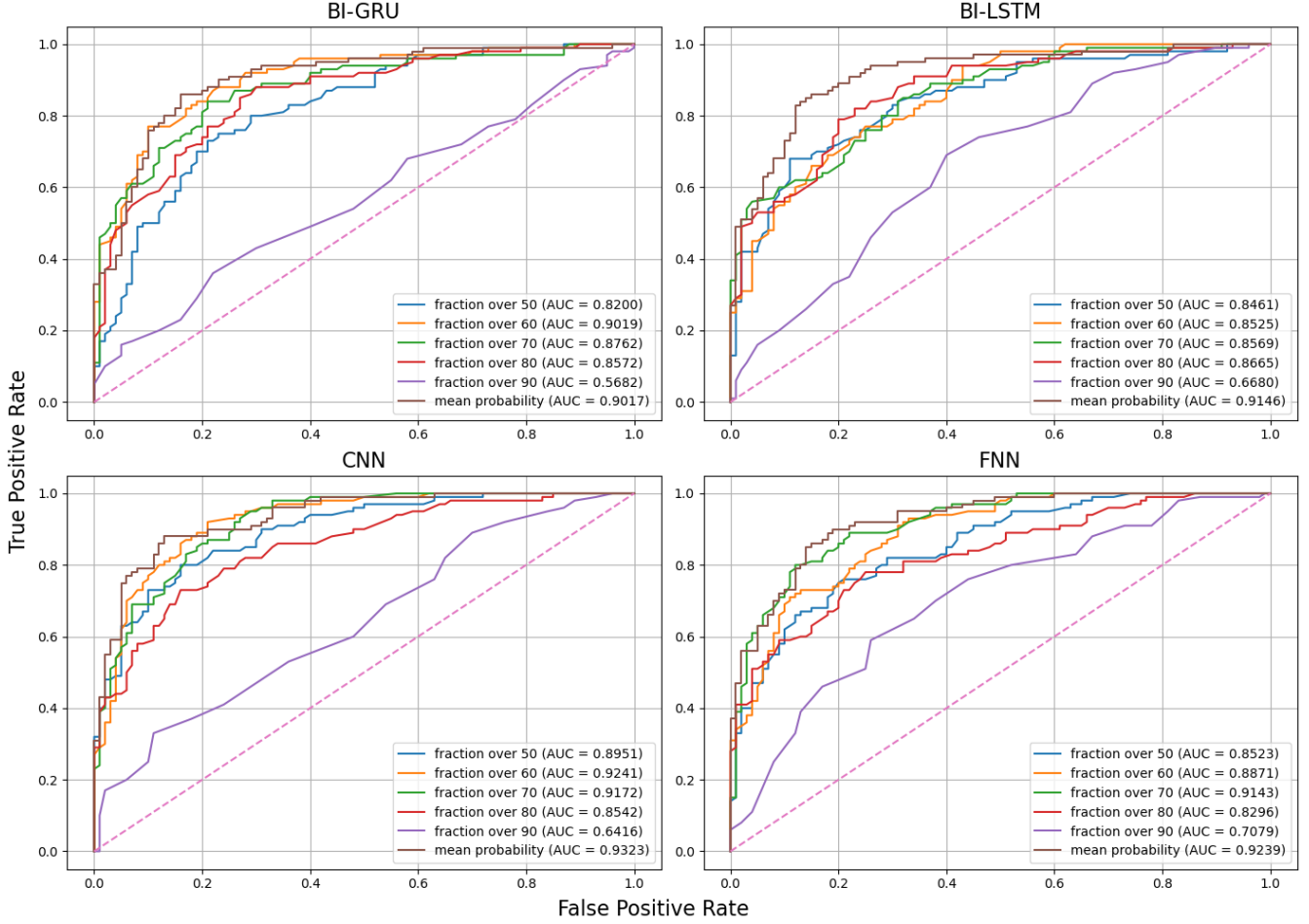


Figure 13: ROC curves and corresponding AUC scores for the aggregation strategies classifying simulated samples with ctDNA fractions of 0.0 and 0.004.

Subsequently, the selected architectures were evaluated on the simulated sample-level test dataset described in Section 2.4. Consistent with the fragment-level evaluation, all deep learning architectures achieved substantially stronger sample-level predictive performance than the logistic regression baseline. The logistic regression results are therefore reported in Appendix D and E, while the following analysis focuses on the relative differences between the deep learning architectures. Here, the results obtained when classifying samples with ctDNA fractions of 0.0 and 0.004 using the different aggregation strategies are shown in Figure 13.

The relative ranking of the evaluated aggregation strategies varied somewhat across architectures. Nevertheless, the mean probability aggregation strategy consistently achieved the highest or near-highest AUC values for all architectures, with the BI-GRU architecture representing the only case in which another strategy marginally outperformed it. In contrast, the fraction-over-90 strategy consistently exhibited substantially lower performance than the remaining aggregation strategies, suggesting that the fragment-level predictions produced by the architectures are generally not highly discriminative. Based on its consistently strong performance, mean probability was selected as the aggregation strategy for the remaining sample-level evaluations.

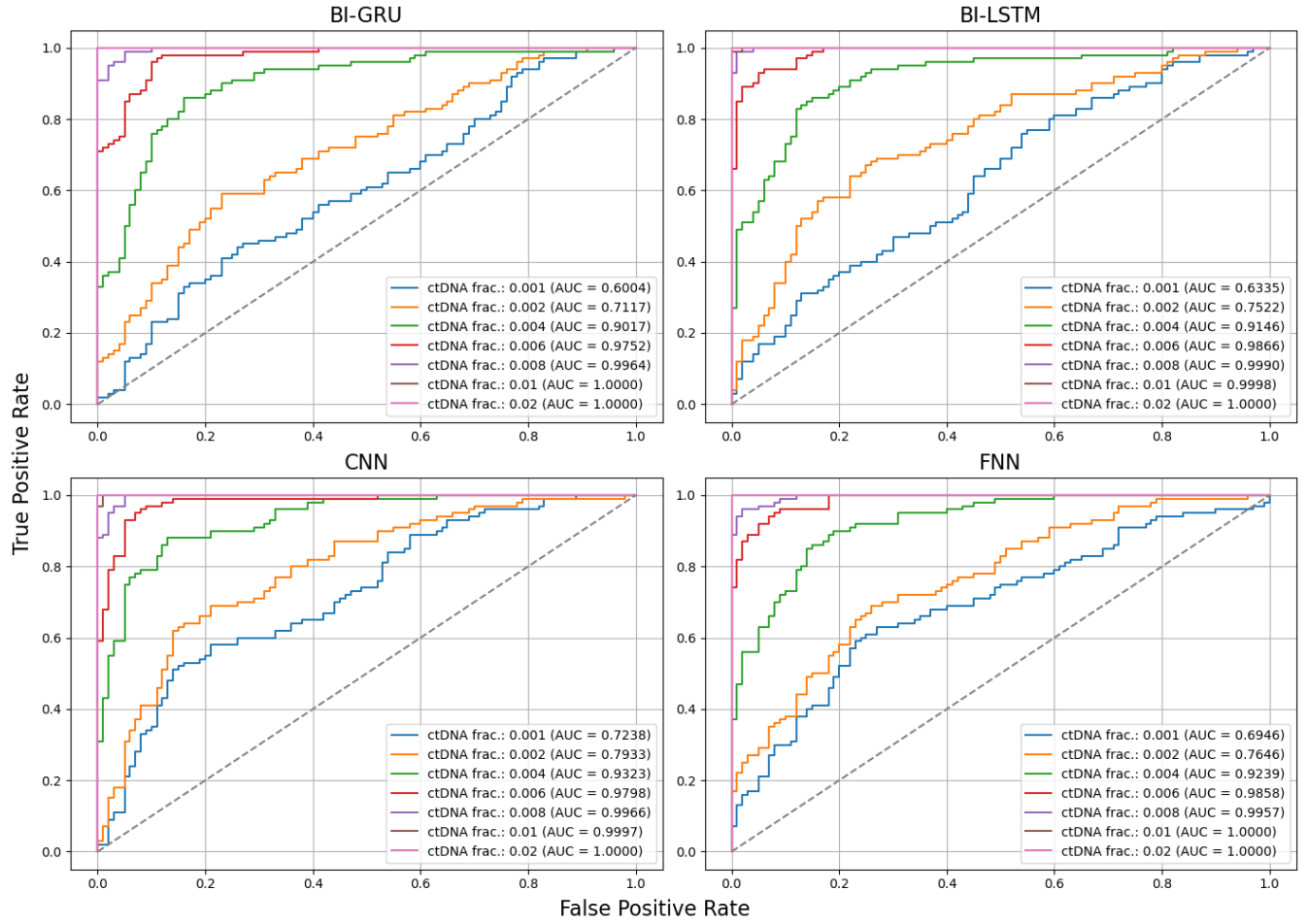


Figure 14: ROC curves and corresponding AUC scores for sample-level classification across varying ctDNA fractions using mean probability as aggregation strategy.

The results obtained when classifying simulated samples with the remaining ctDNA fractions against simulated samples without ctDNA using mean probability as aggregation strategy are shown in Figure 14. All architectures achieved near-perfect classification performance for samples with ctDNA fractions of 2% and 1%, while consistently strong classification performance was observed down to ctDNA fractions of 0.4%. However, classification performance deteriorated substantially below this level, particularly for samples containing only 0.1% ctDNA. This observation is especially relevant, as early-stage cancer samples may exhibit ctDNA fractions below 0.1%. When comparing the architectures at low ctDNA fractions, a relatively consistent performance ranking emerged. The CNN architecture achieved the highest AUC values across the lowest ctDNA fractions, followed by the FNN, BI-LSTM, and BI-GRU architectures. Notably, this ranking differs from that observed during fragment-level evaluation, where the BI-LSTM architecture achieved the highest fragment-level AUC. Intuitively, one might expect the fragment-level ranking to persist in the sample-level setting. However, the present results suggest that fragment-level AUC alone may not fully capture the suitability of a model for sample-level ctDNA detection. However, consistent with expectation the performance differences between architectures became substantially more pronounced in the sample-level setting than during fragment-level evaluation.

3.3 SHAP analysis

To investigate how the implemented fragmentomic features influenced model predictions, a SHAP analysis was performed for the selected model variant from each architecture. The mean absolute SHAP values associated with the nucleotide positions within the end motifs are shown in Figure 15.

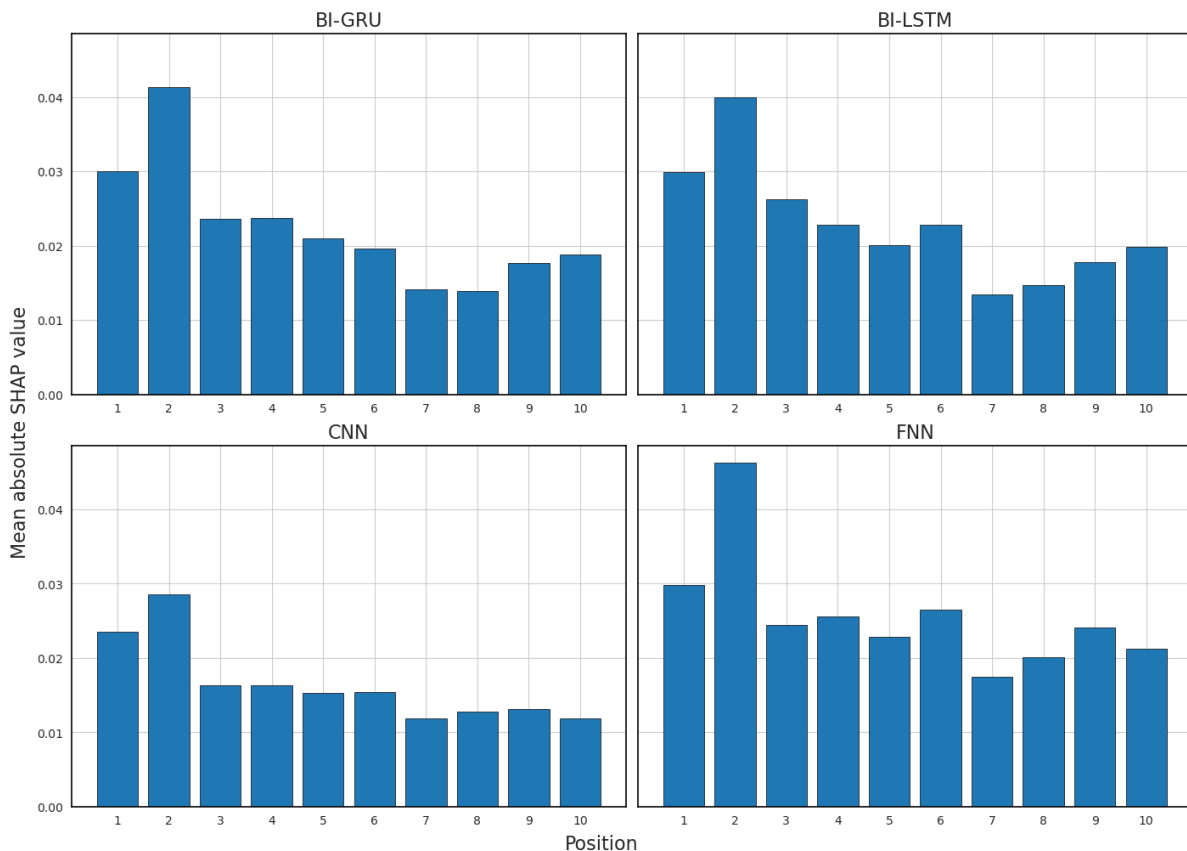


Figure 15: Mean absolute SHAP value for the different nucleotide positions in the end motifs.

Across all architectures, nucleotide positions 1 and 2 exhibited the strongest influence on the model predictions, with position 2 consistently showing the highest mean absolute SHAP value. For the remaining positions, the influence generally decreased with increasing distance from the fragment end. Despite the decrease in influence associated with an increase in nucleotide position, nucleotides beyond the fourth position still contributed substantially to the total influence assigned to the end motifs. Previous studies have primarily focused on the four terminal nucleotides when implementing fragment end motifs for ctDNA detection [18, 20, 26]. However, the above observations suggest that it might be beneficial to incorporate a larger sequence context when implementing end motifs for ctDNA detection, although the strongest influence still appears to originate from the nucleotides closest to the fragment end.

Interestingly, the CNN exhibited a flatter distribution of influence across the nucleotide positions. This suggests that its predictions were influenced more broadly by all the nucleotides across the end motif sequence, rather than being dominated by specific nucleotide positions. This is likely due to

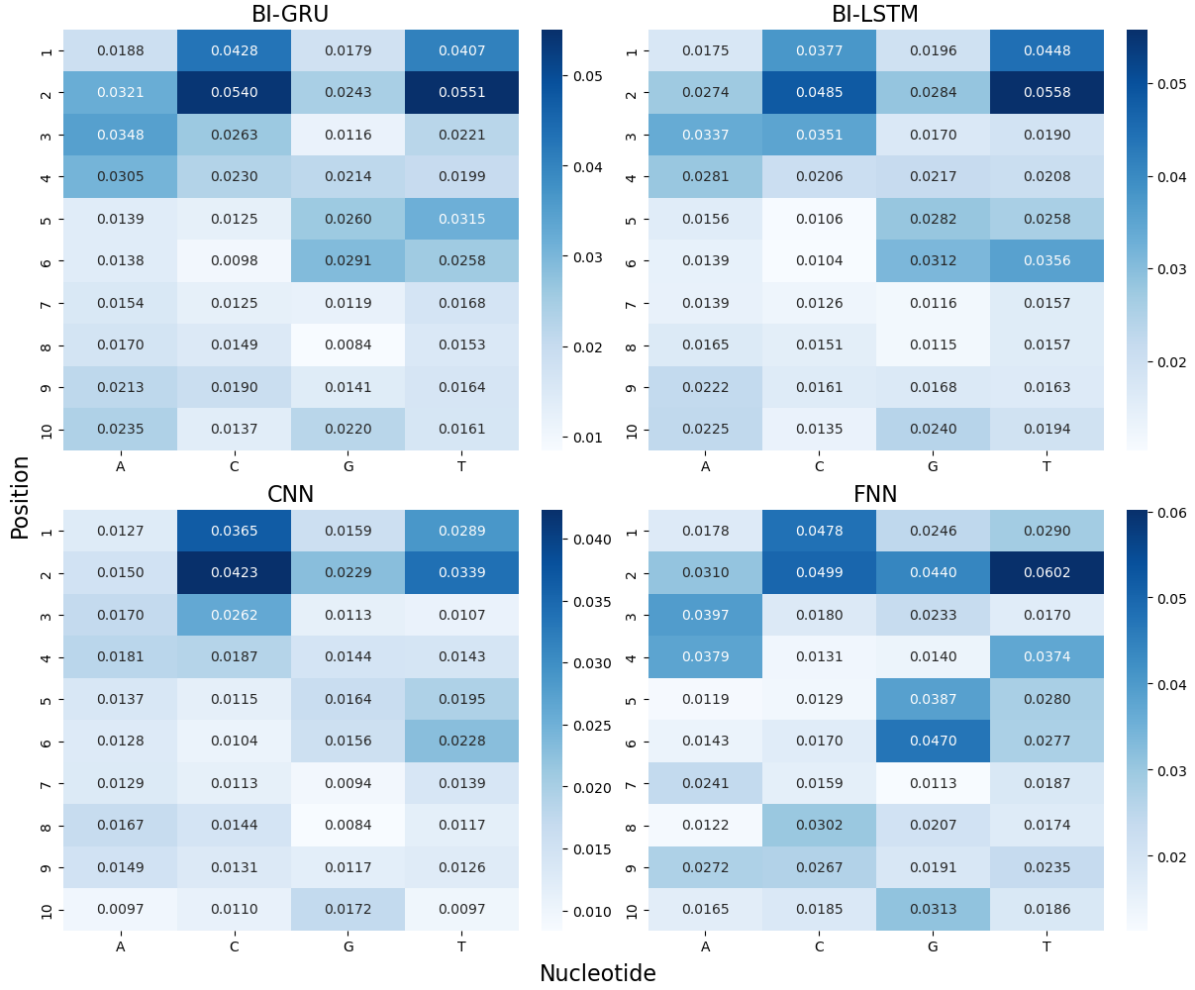


Figure 16: Mean absolute SHAP value for the different nucleotide values for each nucleotide position in the end motifs.

the way convolutional kernels function as local motif detectors. Since each kernel is applied across the entire sequence, a CNN primarily learns important short nucleotide patterns independently of their exact position within the end motif. In this case, however, the CNN layer was adapted to preserve as much positional information as possible. As mentioned, this was done by excluding pooling and using padding. This adaptation may explain why positions 1 and 2 still exhibited comparatively high influence. In contrast, the RNN and FNN architectures rely more directly on explicit positional nucleotide values and sequential relationships between positions, resulting in stronger positional importance.

To investigate whether the influence of the nucleotide positions were associated with specific nucleotide values, heatmaps showing the mean absolute SHAP value for each combination of nucleotide-value and position were created for all architectures, as shown in Figure 16. Across all architectures, positions 1 and 2 exhibited comparatively high influence for nucleotides C and T, suggesting that the models consistently assigned substantial importance to these combinations of nucleotide-value and position. This was the only consistent pattern observed across all architectures.

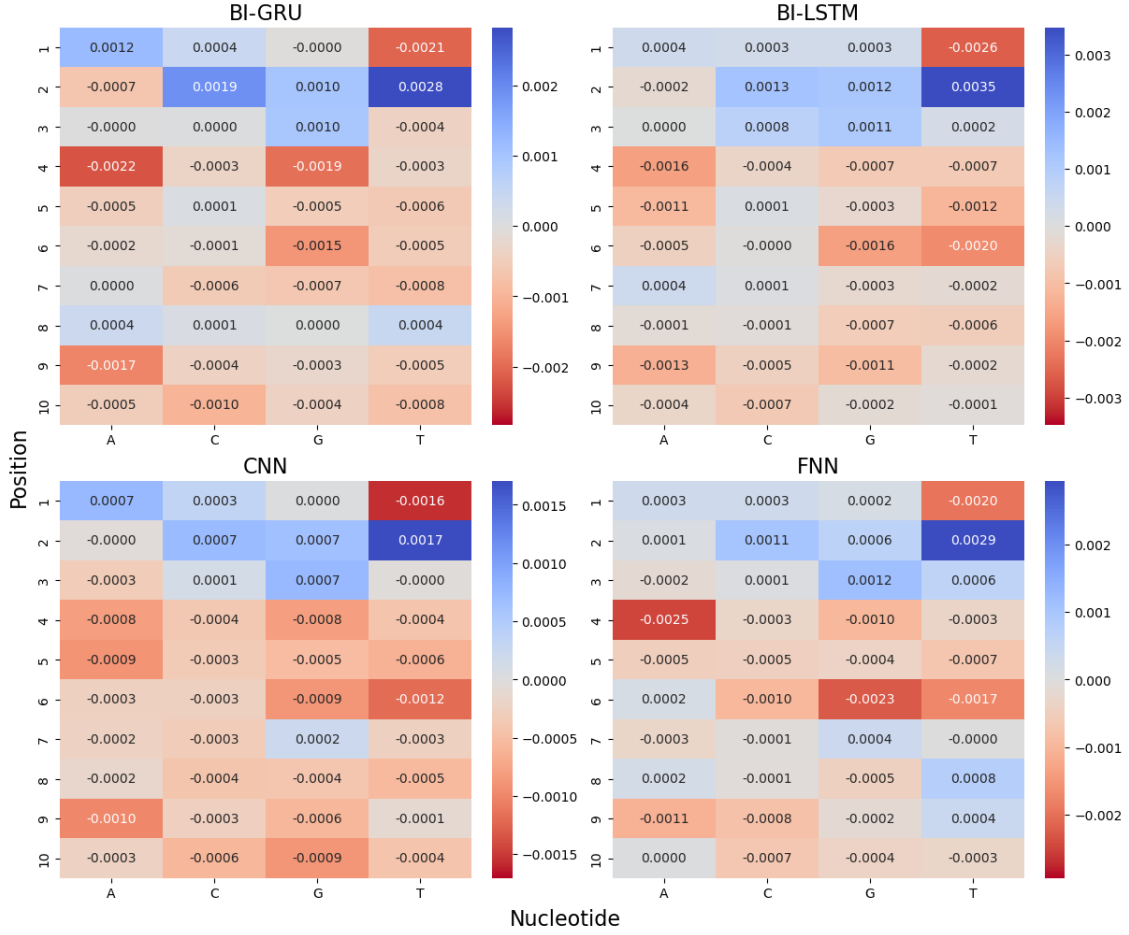


Figure 17: Mean SHAP values for the different nucleotide values for each nucleotide position in the end motifs.

In addition, the BI-GRU, BI-LSTM, and FNN architectures exhibited increased influence associated with nucleotide A at position 3.

However, interpretation of the biological implications of these patterns based solely on mean absolute SHAP values is difficult, as the direction of the effect is not reflected. Therefore, additional heatmaps illustrating the mean signed SHAP value for each combination of nucleotide-value and position were generated. These results are shown in Figure 17. The directional SHAP values shown in Figure 17 enabled further interpretation of the patterns identified in Figure 16. Nucleotides C and T at position 2 contributed positively to the predicted ctDNA probability across all architectures. Furthermore, nucleotide T at position 1 generally contributed negatively to the predicted ctDNA probability. The consistency of these patterns across all architectures suggests that they may reflect underlying biological signals associated with ctDNA. Finally, for the BI-GRU, BI-LSTM, and FNN architectures, nucleotide A at position 4 was associated with a negative contribution to the predicted ctDNA probability.

Finally, the distribution of influence between fragment length and end motif was investigated. Here, the fraction of the total feature influence associated with each feature was calculated using the

Architecture	Fragment length	End motif
FNN	0.69	0.31
CNN	0.76	0.24
BI-LSTM	0.71	0.29
BI-GRU	0.71	0.29

Table 7: Fraction of the total feature influence associated with the implemented features for each architecture.

mean absolute SHAP values and is shown in Table 7. Across all architectures, fragment length accounted for the majority of the total feature influence, indicating that fragment length constituted a comparatively stronger predictive signal than the end motifs within the implemented models. These observations are consistent with the findings from Section 3.1.1, where fragment length demonstrated substantially stronger standalone predictive performance than end motifs. The CNN architecture exhibited the strongest relative dependence on fragment length among the evaluated architectures.

4 Discussion

The primary objective of this thesis was to develop models for fragment-level ctDNA detection. More precisely, the thesis focused on whether it would be possible to develop a relatively simple deep learning architecture which was capable of effectively identifying ctDNA at fragment-level. The results demonstrated that all the architectures introduced in the thesis outperformed the logistic regression baseline model when distinguishing ctDNA fragments from non-tumor-derived cfDNA fragments. However, only relatively small differences in fragment-level predictive performance was found between the introduced architectures. Substantially larger differences emerged in the sample-level setting, particularly at low ctDNA fractions. Notably, the CNN architecture achieved the strongest performance for low-fraction sample-level classification despite not achieving the highest fragment-level AUC. Furthermore, the SHAP analysis indicated that fragment length constituted the dominant predictive signal across all architectures, while end motif information provided complementary predictive value.

In the following section, the obtained results will be discussed in relation to previous research. Additionally, their possible implications for further research and their biological implications will be investigated.

4.1 Feature integration and architectural complexity for fragment-level ctDNA detection

The results from the initial hyperparameter optimization demonstrated that the single-feature models based exclusively on fragment length achieved substantially stronger fragment-level predictive performance than the corresponding models based exclusively on end motifs. Furthermore, the SHAP analysis showed that model predictions for all the selected model variants were influenced

predominantly by the fragment length feature. Together, these observations suggest that fragment length may constitute a stronger predictive signal for ctDNA detection than end motifs within the present dataset. However, caution is required when interpreting feature importance as direct evidence of underlying biological signal strength, as model behavior does not necessarily reflect biological causality. The comparatively lower contribution of end motifs could also reflect limitations in the implemented modeling strategies rather than weaker biological relevance of the feature itself. However, the consistent preference for fragment length across all architectures strengthens the interpretation that fragment length contains comparatively strong predictive information for fragment-level ctDNA detection.

The initial hyperparameter optimization further demonstrated that the multimodal FNN variants achieved substantially stronger fragment-level predictive performance than the corresponding single-feature variants. Consequently, to reduce the total number of model variants, only multimodal variants were implemented for the remaining architectures. As a result, it remains unknown whether the end motif modeling approaches introduced in the remaining architectures would have achieved superior performance in a single-feature setting. However, this appears unlikely given the comparatively limited fragment-level performance improvement observed for the multimodal variants when introducing more advanced end motif modeling architectures. Furthermore, the improvement in fragment-level predictive performance observed when transitioning from the single-feature end motif FNN to the multimodal FNN was among the largest architectural effects observed. Finally, the SHAP analysis demonstrated that fragment length constituted the dominant feature influencing model predictions across all evaluated architectures. Together, these observations suggest that removal of the fragment length feature would likely reduce, rather than improve, predictive performance.

Although fragment length constituted the dominant feature influencing model predictions, the choice of fragment length preprocessing strategy appeared to have only limited impact on fragment-level predictive performance. In particular, representing fragment length using one-hot encoding did not provide substantial performance improvements compared with normalized fragment length, despite more explicitly reflecting the discrete nature of fragment lengths. This observation may suggest that the predictive signal associated with fragment length can be captured effectively even when represented as a continuous feature. In contrast, the choice of end motif representation had a more pronounced effect on predictive performance. Across all evaluated architectures, the combined end motif setting consistently outperformed the individual end motif setting. This suggests that the two fragment ends contain complementary predictive information that can be utilized jointly to improve ctDNA detection performance.

An additional observation emerging from the initial hyperparameter search is that changes related to feature representation and feature integration generally had a larger effect on fragment-level predictive performance than the choice of sequence-modeling architecture. In particular, the transition from single-feature to multimodal modeling and the choice of combined rather than individual end motif setting resulted in substantially larger improvements than when transitioning between the FNN, CNN, BI-LSTM, and BI-GRU architectures.

The final model evaluation demonstrated a similar pattern as the initial hyperparameter search. Despite the substantial differences in architectural complexity, only minimal differences in fragment-level predictive performance were observed between the evaluated deep learning architectures. These differences were so minimal that they could have been caused by the relatively high fluctuation in performance during training, as shown in Figure 12, rather than reflecting meaningful differences in predictive capability between the architectures. This further supports the argument that changes related to feature representation and feature integration generally have a larger effect on fragment-level prediction performance than increasing sequence-modeling architectural complexity. Nonetheless, all the implemented deep learning architectures outperformed the logistic regression baseline model at both fragment-level and sample-level prediction. This indicates that nonlinear deep learning approaches may provide advantages over conventional linear machine learning models for both fragment-level and sample-level ctDNA detection. However, the achieved fragment-level performances remained moderate, suggesting that reliable classification of individual fragments remains challenging. Consequently, the primary value of the introduced architectures may not lie in confident classification of individual fragments, but rather in their ability to aggregate moderate fragment-level prediction into reliable sample-level predictions.

Together, the above observations suggest that fragment-level ctDNA detection depends strongly on how the fragmentomic features are represented and integrated within the model framework. In particular, modifications affecting feature utilization, such as multimodal modeling and combined end motif setting, appeared to have a greater effect on predictive performance than increasing sequence-modeling architectural complexity. Furthermore, they indicate that meaningful ctDNA-associated information can be extracted from individual fragments using relatively simple deep learning architectures when combined with appropriate strategies for representing and integrating fragmentomic information. However, the achieved fragment-level predictive performance remained only moderate across all architectures, indicating that reliable classification of individual fragments remains challenging. Consequently, the primary value of fragment-level models may lie in their ability to aggregate moderate fragment-level predictions into robust sample-level predictions.

4.2 Sample-level prediction performance

The sample-level evaluation revealed a substantially different pattern from that observed in the fragment-level setting. In particular, clearer performance differences emerged between the deep learning architectures when evaluating classification performance on simulated samples with low ctDNA fractions. Here, the CNN architecture consistently achieved the strongest performance, followed by the FNN, BI-LSTM, and BI-GRU architectures, respectively. This ranking differs from the ranking observed in the fragment-level evaluation, further suggesting that the minimal differences in fragment-level performance may partly reflect variability during training rather than meaningful differences in fragment-level prediction performance. Consequently, the stronger separation between architectures in the sample-level setting cannot simply be interpreted as an amplification of the fragment-level differences. A possible explanation for this stronger separation is that differences in the distribution of fragment-level predictions between the architectures influence the aggregated sample-level predictions. Since the selected aggregation strategy was based on the mean fragment-level prediction, the resulting sample-level prediction is sensitive to highly discriminative predictions. Consequently, architectures producing more extreme predictions

could potentially achieve improved sample-level classification performance without affecting the fragment-level AUC. This is possible as the AUC is only affected by the ranking of the predictions and not their extremities. Nevertheless, the CNN architecture outperformed the remaining deep learning architectures when evaluated on simulated samples with a low ctDNA fraction. This suggests that the CNN architecture may be particularly well suited for applications involving early-stage cancer detection, where ctDNA fractions are expected to be very low. This corresponds well with the architectural reasoning presented in Section 2.2.2, where the CNN architecture was proposed as an effective compromise between model complexity and prediction performance for short end motif modeling.

To further evaluate the significance of the sample-level results obtained by the CNN architecture, comparison with existing approaches from the liquid biopsy field is relevant. However, establishing direct comparisons between fragmentomic-based methods remains difficult for a number of reasons. Firstly, fragmentomic-based liquid biopsy for early-stage cancer detection is still a relatively new research area, and no standardized benchmark datasets or evaluation frameworks have yet been established. Secondly, many previous fragmentomic studies have primarily focused on demonstrating proof-of-concept performance across all cancer stages rather than specifically evaluating low ctDNA fraction settings, as was the case with the DELFI method [25]. Among studies that have investigated low ctDNA fraction detection, simulated dilution experiments are commonly used. Typically, these studies generate low-fraction samples by diluting samples with estimated high ctDNA fractions using fragments from negative control samples. However, such approaches do not provide exact knowledge of the resulting ctDNA fraction [54]. Finally, fragmentomic characteristics are strongly influenced by sequencing protocols and laboratory procedures and therefore vary between dataset. Furthermore, the difficulty of the prediction task depends on the included cancer types and control populations in the dataset. Consequently, fair comparison between fragmentomic methods generally requires evaluations based on the same dataset [55]. These limitations prohibit comparison with fragmentomic-based methods. However, comparison with methods from the broader liquid biopsy field may still provide useful context regarding the current state of early-stage cancer detection. In this regard, the ichorCNA method introduced by Adalsteinsson et al. established an important baseline by achieving a sensitivity of 0.95 and specificity of 0.91 for samples with ctDNA fractions down to 0.03 [56]. Notably, all architectures implemented in this thesis achieved near-perfect separation between samples with ctDNA fractions of 0.01 and negative samples. Furthermore, the CNN architecture maintained strong classification performance at ctDNA fractions as low as 0.006, where it achieved a sensitivity of 0.96 and specificity of 0.91. Although these results cannot be compared directly with ichorCNA due to differences in datasets and evaluation strategies, they nevertheless suggest that the proposed architectures may retain meaningful predictive capability at ctDNA fractions below the reported detection limit of ichorCNA. Importantly, the present results were obtained using simulated low fraction samples. Validation on real low ctDNA fraction samples is therefore necessary before conclusions regarding clinical applicability can be drawn. Such validation represents an important direction for future research.

The above observations suggest that architectural differences become substantially more important in the sample-level setting than in the fragment-level setting, particularly for classification of samples with low ctDNA fractions. In particular, the CNN architecture consistently demonstrated strong low-fraction classification performance, indicating that CNN-based modeling of short end

motif sequences may be especially promising for early-stage cancer detection. However, further validation on real low-ctDNA-fraction samples is necessary before conclusions regarding clinical applicability can be established.

4.3 SHAP analysis

The SHAP analysis provided insight into how the implemented fragmentomic features influenced the predictions of the different model architectures. As discussed previously, fragment length constituted the dominant predictive feature across all architectures, although end motif information consistently contributed complementary predictive value. Further analysis of the end motif contributions provided additional insight into the sequence context relevant for fragment-level ctDNA detection. Previous studies have primarily focused on the four terminal nucleotides when implementing end motif-based ctDNA detection methods. However, the results presented in Section 3.3 indicated that all nucleotide positions included in the present analysis contributed meaningfully to the model predictions. These observations suggest that incorporating a larger sequence context than the traditionally used four terminal nucleotides may be beneficial for end motif modeling. It could therefore be relevant for future studies to investigate even longer end motif representations in order to determine how far the predictive influence of the sequence context extends. The SHAP analysis further demonstrated that the comparatively high influence associated with nucleotide positions 1 and 2 was driven primarily by the nucleotide values C and T. In particular, T at positions 1 and 2 exhibited a consistent directional effect on model predictions across all architectures. Importantly, these observations reflect how the implemented models utilized the fragmentomic features and do not by themselves establish underlying biological mechanisms. Nevertheless, the consistency of these patterns across all evaluated architectures suggests that they may reflect biologically relevant fragmentomic characteristics associated with ctDNA.

Since the CNN architecture showed comparatively strong sample-level performance, it would be interesting to perform a further analysis of its learned convolutional kernels. As mentioned in section 3.3, the kernel structure of the CNN architecture makes it less dependent on the specific nucleotide positions compared to the RNN and FNN architectures. Consequently, the SHAP analysis of the end motif influence for the CNN architecture becomes less informative. Since the convolutional kernels from the CNN architecture function as local motif detectors, a direct analysis of the learned convolutional filters could provide additional biological and methodological insight into which local sequence patterns are considered important by the CNN architecture for ctDNA prediction. Such analyses have previously proven useful in genomic CNN models for identifying biologically meaningful sequence motifs.

5 Conclusion

The primary objective of this thesis was to develop and evaluate models for fragment-level ctDNA detection using a limited set of fragmentomic features. To investigate this objective, multiple deep learning architectures implementing fragment length and end motif information were developed, optimized, and evaluated at both fragment-level and sample-level.

All implemented deep learning architectures outperformed the logistic regression baseline model,

indicating that nonlinear deep learning approaches are capable of capturing predictive ctDNA patterns beyond those available to conventional linear machine learning methods. However, only moderate fragment-level predictive performance was achieved across all architectures. This indicates that meaningful ctDNA-associated information can be extracted from individual fragments using relatively simple deep learning architectures, but reliable classification of individual fragments remains challenging.

The results further showed that decisions related to feature representation and feature integration generally had a greater influence on fragment-level predictive performance than the choice of sequence-modeling architecture. In particular, multimodal modeling and the combined end motif setting consistently improved predictive performance, while only minor differences were observed between the evaluated deep learning architectures at fragment-level. These findings suggest that effective utilization of fragmentomic information may be more important than highly complex model architectures for fragment-level ctDNA detection.

Although fragment-level predictive performance remained moderate, aggregation of fragment-level predictions enabled accurate sample-level classification of simulated samples across a range of ctDNA fractions. Interestingly, the relatively small performance differences observed between architectures at fragment-level became substantially more pronounced in the sample-level setting, particularly for samples with low ctDNA fractions. This suggests that while architectural choices may have limited influence on fragment-level prediction performance, they may play a more important role in determining how effectively fragment-level signals are translated into sample-level predictions. In this setting, the CNN architecture achieved the strongest overall performance, particularly for samples containing low ctDNA fractions. However, the sample-level evaluation was based on simulated samples generated from fragment-level data and should therefore primarily be interpreted as a proof of concept demonstrating the potential utility of fragment-level predictions for sample-level ctDNA detection. Validation on independent real-world datasets is required before conclusions regarding clinical applicability can be drawn.

The SHAP analysis further provided insight into how the implemented models utilized the fragmentomic features. Fragment length was consistently identified as the dominant predictive feature, while end motif information contributed complementary predictive value across all architectures. Furthermore, the analysis suggested that sequence information beyond the four terminal nucleotides may contain relevant predictive information, indicating a potential direction for future fragmentomic research.

Overall, the findings of this thesis demonstrate that fragment-level modeling represents a promising and largely unexplored direction within fragmentomic-based ctDNA detection. While reliable classification of individual fragments remains challenging, the results suggest that aggregating predictions from moderately accurate fragment-level models may constitute a viable strategy for sample-level ctDNA detection. Future work should focus on validating these findings on independent clinical datasets.

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Appendix

Appendix A

Heatmap showing the mean predictive performance from the initial hyperparameter search, indicated by mean validation AUC, for interactions between hyperparameters for the CNN architecture.

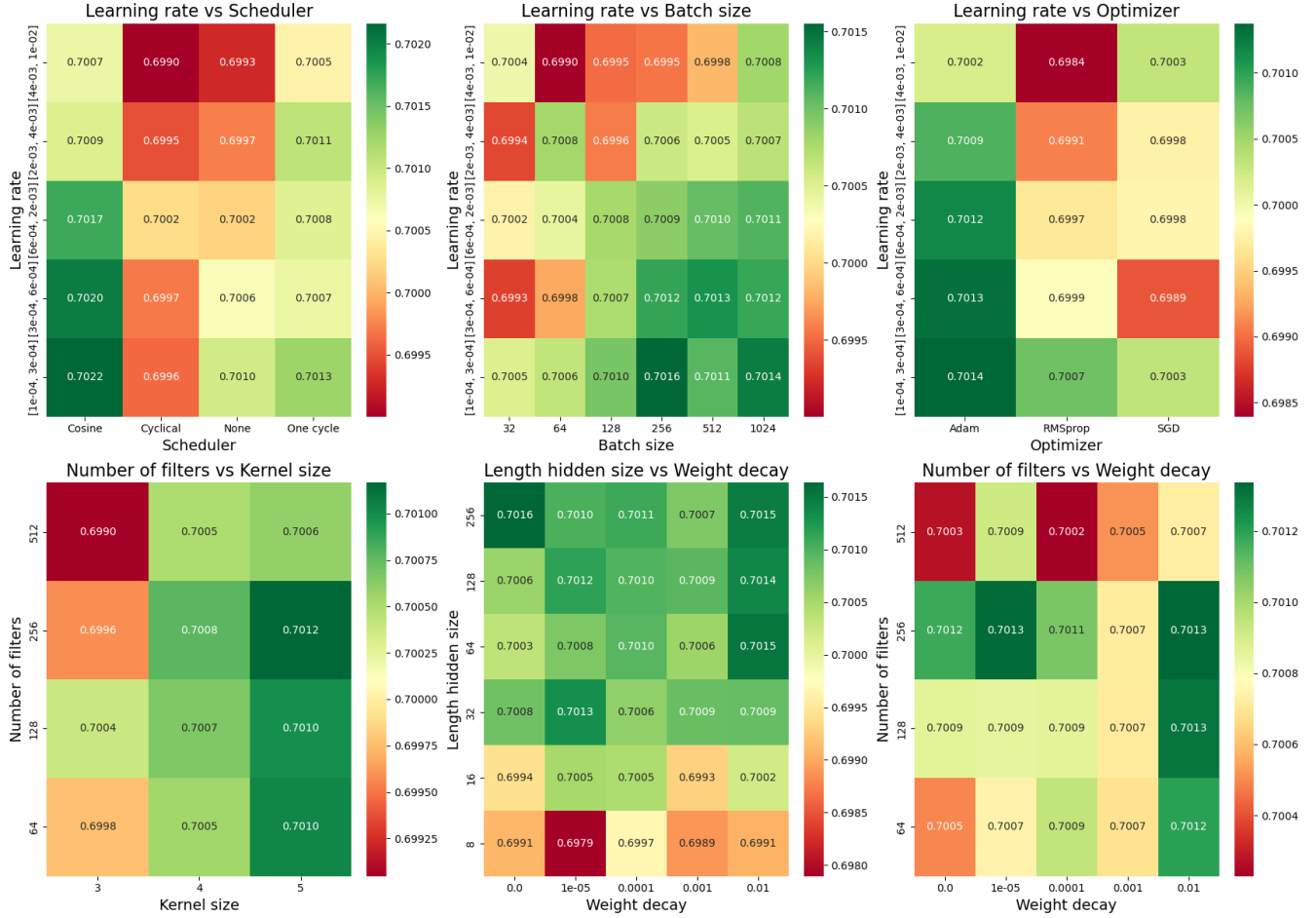


Figure 18: The effect of different hyperparameter interactions on validation AUC for the CNN architecture. For each interaction the mean validation AUC for all configurations implementing the corresponding interaction is shown.

Appendix B

Heatmap showing the mean predictive performance from the initial hyperparameter search, indicated by mean validation AUC, for interactions between hyperparameters for the LSTM architecture.

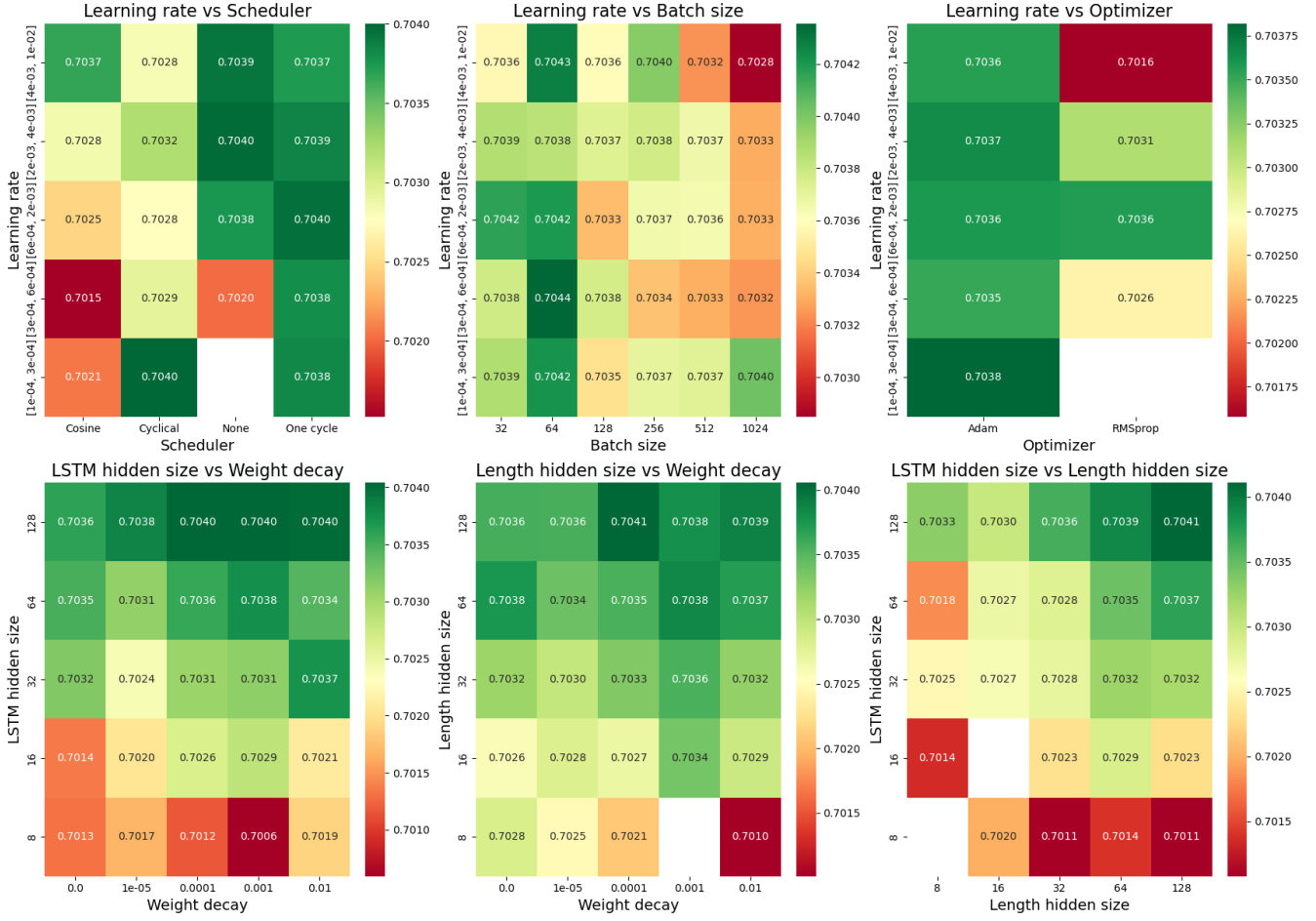


Figure 19: The effect of different hyperparameter interactions on validation AUC for the LSTM architecture. For each interaction the mean validation AUC for all configurations implementing the corresponding interaction is shown.

Appendix C

Heatmap showing the mean predictive performance from the initial hyperparameter search, indicated by mean validation AUC, for interactions between hyperparameters for the GRU architecture.

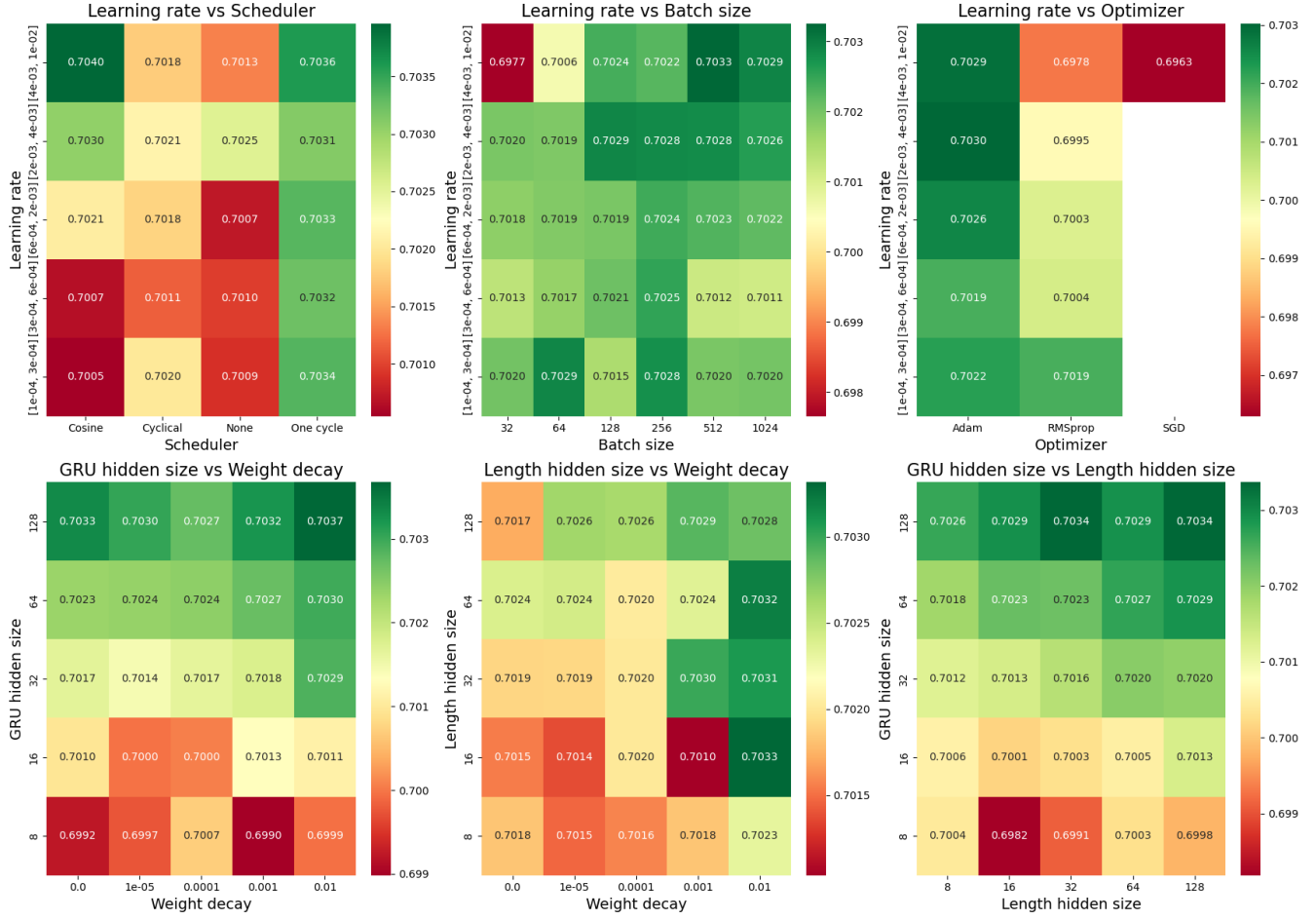


Figure 20: The effect of different hyperparameter interactions on validation AUC for the GRU architecture. For each interaction the mean validation AUC for all configurations implementing the corresponding interaction is shown.

Appendix D

ROC curves and corresponding AUC score for the different aggregation strategies for the logistic regression classifying simulated samples with ctDNA fraction of 0.0 and 0.04.

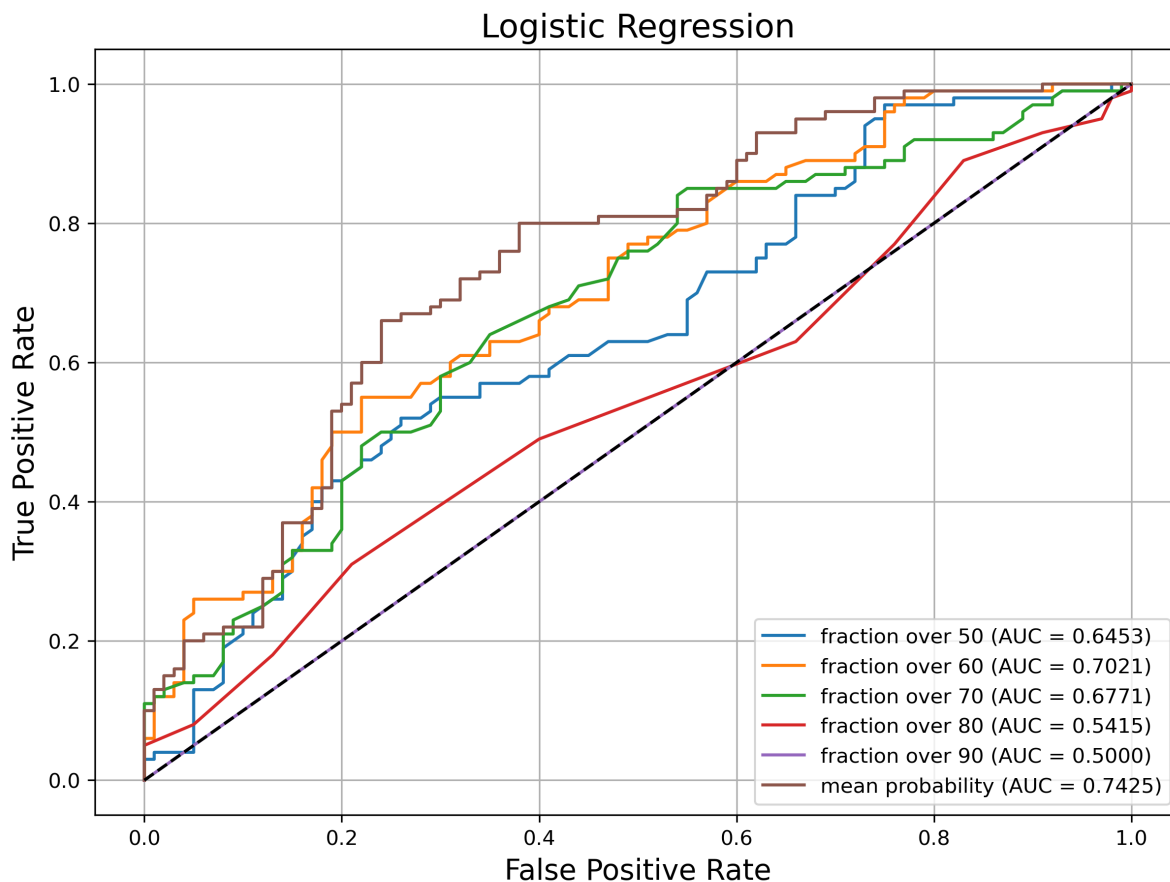


Figure 21: ROC curves and corresponding AUC scores for the aggregation strategies classifying simulated samples with ctDNA fractions of 0.0 and 0.004.

Appendix E

ROC curves and corresponding AUC score when classifying simulated samples with the varying ctDNA fractions against simulated samples without ctDNA using mean probability as aggregation strategy for logistic regression.

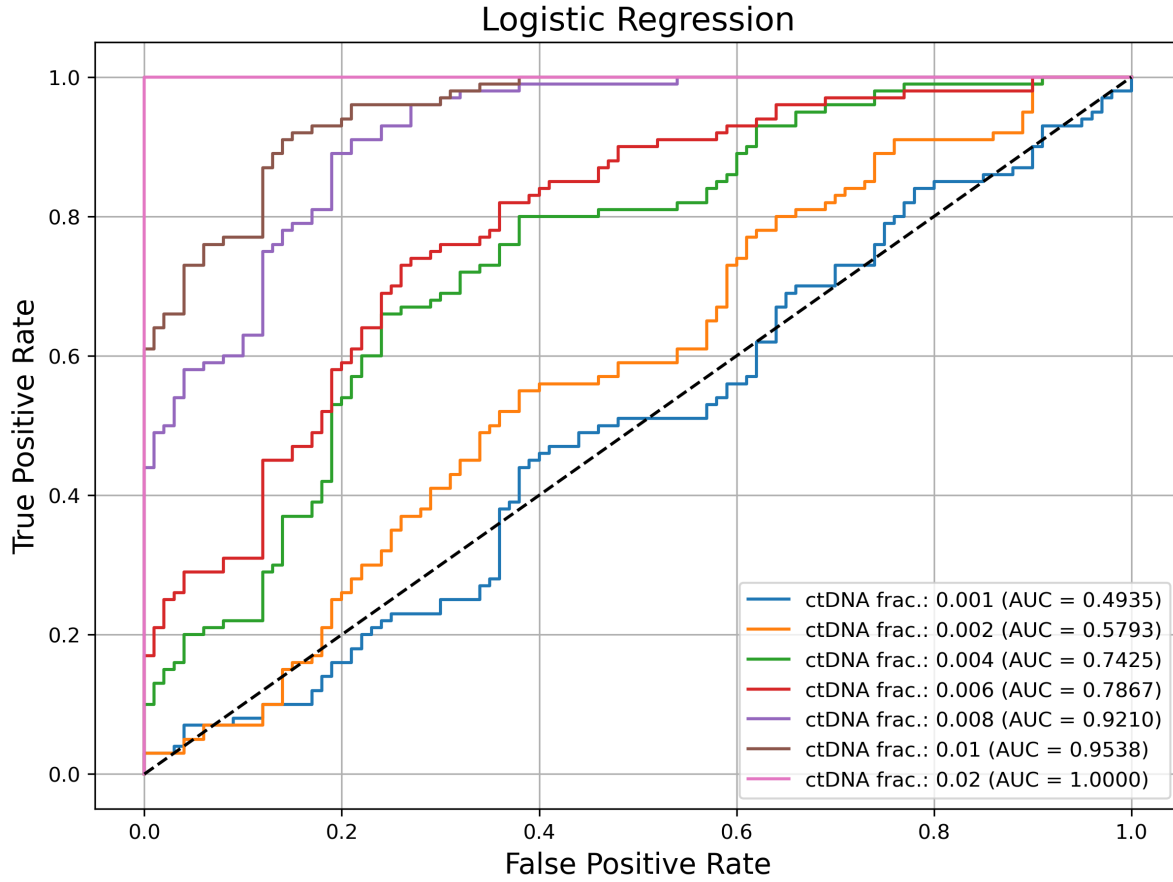


Figure 22: ROC curves and corresponding AUC scores for sample-level classification across varying ctDNA fractions using mean probability as aggregation strategy.